

Global, regional, and national burden of tuberculosis and multidrug-resistant tuberculosis by HIV status, 1990–2023: a systematic analysis for the Global Burden of Disease Study 2023



GBD 2023 TB HIV Collaborators*

Summary

Background Tuberculosis (TB) is the leading global cause of death from a single infectious agent. Recent reductions in global health funding have threatened TB control, making comprehensive assessment of TB, HIV-related TB, and drug-resistant TB burdens before these disruptions essential for shaping effective responses. The WHO End TB Strategy sets targets of a 95% reduction in TB deaths and a 90% reduction in TB incidence between 2015 and 2035. Using results from the Global Burden of Diseases, Injuries, and Risk Factors Study (GBD) 2023, this study aims to assess the burden of TB and multidrug-resistant TB (MDR-TB) across 204 countries and territories, and to evaluate progress towards the WHO End TB incidence and mortality targets.

Methods We quantified TB mortality using the Cause of Death Ensemble modelling platform with global vital registration, surveillance, verbal autopsy, and minimally invasive tissue sampling data. For TB morbidity estimation, we simultaneously modelled incidence, prevalence, and mortality by age and sex using DisMod-MR 2.1. A population attributable fraction (PAF) approach was applied to stratify morbidity and mortality estimates by HIV and drug-resistance status. We also calculated disability-adjusted life-years (DALYs) as the sum of years of life lost and years lived with disability. For the risk factor analysis, a comparative risk assessment framework was used and PAFs were derived for alcohol use, smoking, and high fasting plasma glucose to determine the proportion of TB burden associated with these risk factors.

Findings In 2023, there were an estimated 9·11 million (95% uncertainty interval 8·04–10·3) incident cases of all-form TB, 1·22 million (0·98–1·49) deaths, and 54·6 million (43·8–65·5) DALYs globally. HIV-related TB comprised 781 000 (690 000–879 000) incident cases and 210 000 (142 000–279 000) deaths, contributing 11·0 million (7·56–14·3) DALYs. MDR-TB accounted for 466 000 (198 000–1 080 000) incident cases, 102 000 (31 700–238 000) deaths, and 3·96 million (1·31–9·01) DALYs. From 2015 to 2023, global all-form TB incidence rates declined by 19·2% (17·8–20·5) and deaths declined by 22·6% (4·7–35·7); declines were larger for drug-susceptible TB than for MDR-TB. Sub-Saharan Africa and south Asia had the highest mortality burdens in 2023; reductions in all-form TB incidence and mortality were uneven between 2000 and 2023, with limited progress in both measures in Latin America and the Caribbean. Removing smoking, alcohol use, and high fasting plasma glucose would reduce global TB deaths to 768 000 (592 000–970 000) and DALYs to 34·9 million (27·8–43·8) in 2023; MDR-TB deaths would decrease to 77 200 (23 400–183 000) and DALYs to 3·12 million (1·03–7·29).

Interpretation Global progress towards WHO End TB targets is disparate and fragile. Although many regions achieved meaningful gains, others have stagnated in recent years. The complexity of TB prevention is amplified by divergent MDR-TB trends, the persistent burden of HIV, and growing exposure to modifiable risk factors. Recent volatility in global health financing threatens to further destabilise this vulnerable epidemiological landscape; concerted action is urgently needed to temper disruptions and preserve progress.

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Introduction

Tuberculosis (TB) is a preventable, treatable, and curable disease. However, it remains the leading cause of death from a single infectious agent worldwide, causing more than 1 million deaths annually.¹ Despite ongoing global TB prevention efforts, recent declines in TB incidence

and mortality have stagnated. The stagnation likely reflects a complex set of challenges,² including suboptimal case detection, lengthy and toxic treatment regimens,³ health-system limitations,⁴ and social determinants of TB.⁵ These obstacles have contributed to the rise of multidrug-resistant TB (MDR-TB).⁶ MDR-TB,

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*Collaborators listed at the end of the Article

Correspondence to:
Dr Hmwe H Kyu, Department of Health Metrics Sciences and Institute for Health Metrics and Evaluation, University of Washington, Seattle, WA 98195, USA
hmwekyu@uw.edu

Research in context

Evidence before this study

Several groups, including the WHO Global Tuberculosis Programme and the Global Burden of Diseases, Injuries, and Risk Factors Study (GBD), have generated population-based estimates of tuberculosis (TB) incidence and mortality to monitor progress towards TB elimination targets. However, the global burden of multidrug-resistant tuberculosis (MDR-TB) has been less comprehensively analysed, particularly with respect to its divergence from drug-susceptible TB trends, disability-adjusted life-years (DALYs), and risk-deleted burden. We searched PubMed with the search string “(“tuberculosis” OR “TB”) AND (“drug-resistant” OR “MDR-TB” OR “XDR-TB”) AND (“burden” OR “incidence” OR “mortality” OR “DALY*”) AND (“End TB”))”, with no language restrictions, for publications from Jan 1, 2015, to Sept 15, 2025. Although our search identified recent assessments of the MDR-TB landscape—including global analyses of socioeconomic inequities, paediatric-specific trends, and regional burden analyses—gaps remain. Specifically, no existing studies have quantified years of life lost (YLLs), years lived with disability (YLDs), or DALYs for MDR-TB in the context of HIV; previous research, such as GBD 2016, focused primarily on incidence and mortality rather than these comprehensive disability metrics. These data are urgently needed to contextualise the potential impact of recent global health programme disruptions, which are expected to increase acquired drug resistance and threaten progress towards TB elimination targets.

Added value of this study

Although previous GBD publications on TB reported MDR-TB incidence and mortality, this GBD 2023 report represents a

comprehensive effort to provide globally comparable MDR-TB burden estimates inclusive of YLLs, YLDs, and DALYs. These metrics have not been previously quantified for MDR-TB within the context of HIV. Furthermore, we identified a gap in globally comparable data regarding country-level progress towards WHO End TB milestones, specifically concerning the differential between MDR-TB and drug-susceptible TB. For the first time, we report risk-deleted estimates of MDR-TB mortality and DALYs, quantifying the burden that would remain if all three evaluated risk factors (smoking, alcohol use, and high fasting plasma glucose) were removed. Together, these innovations offer the most comprehensive assessment to date of the MDR-TB burden and its contribution to the global TB epidemic, thereby providing timely evidence to guide programmatic and policy responses.

Implications of all the available evidence

Global TB prevention is entering a period of heightened vulnerability. Our estimates reveal substantial progress in the prevention of HIV-associated TB, particularly in sub-Saharan Africa. However, the burden of MDR-TB has not kept pace with gains in drug-susceptible TB, regardless of HIV status. Future reductions in burden hinge on safeguarding the continuity of HIV and TB programmes and accelerating the translation of clinical advances into population-level benefits—imperatives that are now jeopardised by substantial uncertainties in global health financing.

defined as resistance to both isoniazid and rifampicin, carries high mortality rates^{7,8} and more severe, complex post-treatment sequelae.⁹

The need for monitoring MDR-TB trends is heightened by recent disruptions to global health programmes and sharp declines in funding.¹⁰ These programmes have historically supported administrative, clinical, and research capacity while maintaining drug supply chains across high-TB-burden regions.^{11,12} Their dismantling threatens consistent access to anti-TB treatment,¹³ with emerging reports already showing disruptions in drug supply chains, stockouts, and clinic closures.¹⁴ Because many MDR-TB cases arise from interrupted therapy,¹⁵ reductions in programmatic support risk accelerating resistance and contributing to excess preventable deaths from a lack of consistent and extended combination therapies. Comprehensive evaluations of TB burden by drug-resistance status before these disruptions are therefore crucial for shaping strategic responses. At the same time, examining trends in all-form TB remains essential to contextualise the broader implications for TB elimination strategies. Using the Global Burden of Diseases, Injuries, and Risk Factors Study (GBD) 2023

framework, this study aimed to assess the burden of TB and MDR-TB across 204 countries and territories and to systematically evaluate progress towards the WHO End TB Strategy targets, which call for reductions in TB incidence and mortality of 80% and 90%, respectively, by 2030 and 90% and 95%, respectively, by 2035, compared with 2015 levels. This study was conducted as part of the GBD Collaborator Network and in accordance with the GBD Protocol.¹⁶

Methods

Overview

Detailed descriptions of the GBD analytical framework and TB estimation strategy have been published previously.^{1,17–19} Here, we provide a summary of the methods used to estimate TB mortality and morbidity in GBD 2023, including disaggregation by drug resistance and attribution to risk factors. Expanded methodological details and flow diagrams are provided in appendix 1. In line with the GATHER statement,²⁰ input data sources are available through the Global Health Data Exchange and the completed GATHER checklist is provided in appendix 1 (pp 74–75). The University of Washington

See Online for appendix 1

Institutional Review Board approved the GBD study (STUDY00009060). The study is approved until 23 July 2026.

Tuberculosis mortality

The GBD 2023 Cause of Death database incorporated all available global vital registration, mortality surveillance, sample vital registration, and verbal autopsy data from 1980 to 2023.²¹ TB mortality estimates among people without HIV were generated with the Cause of Death Ensemble modelling (CODEm) platform using more than 20 261 site-years of vital registration data, 953 site-years of sample vital registration data, 676 site-years of surveillance data, 2140 site-years of verbal autopsy data, and nine site-years of minimally invasive tissue sample diagnoses. CODEm evaluates a broad set of model specifications with varying functional forms and covariate sets (eg, Healthcare Access and Quality [HAQ] Index, smoking prevalence, and alcohol consumption) and selects the optimal ensemble based on out-of-sample predictive validity.²² Country-specific data sources and citations are available in the Global Health Data Exchange. All raw cause of death data were standardised^{16,23} to account for completeness, reconcile different coding schemes, redistribute deaths from non-specific or intermediate codes to valid underlying causes, and reassign misclassified HIV deaths.^{21,24}

To estimate TB mortality among people living with HIV, we used a population attributable fraction (PAF) approach consistent with previous GBD iterations.^{1,17} Detailed methods of HIV-related TB mortality estimation are provided in appendix 1 (pp 12, 14).

To disaggregate TB mortality by drug-resistance status, we first estimated the proportion of cases that were MDR-TB without being extensively drug resistant (XDR), defined as resistance to the two first-line anti-TB drugs (isoniazid and rifampicin) but no resistance to any fluoroquinolone or any second-line injectable drugs, for all locations and years using spatiotemporal Gaussian process regression. These proportions were stratified by HIV status by use of relative risks from a meta-analysis²⁵ and then combined with the relative risk of death in MDR-TB versus drug-susceptible TB to estimate the fraction of deaths attributable to MDR-TB among people living with and without HIV. We further distinguished XDR-TB (defined as resistance to isoniazid and rifampicin, plus any fluoroquinolone and any second-line injectable drugs) from MDR-TB by estimating the proportion of MDR-TB cases that were XDR-TB at the GBD super-region level and applying corresponding mortality risks.

Tuberculosis morbidity

We jointly estimated all-form TB incidence, prevalence, and cause-specific mortality by age and sex using DisMod-MR 2.1,²⁶ a Bayesian meta-regression tool that ensures internal consistency across epidemiological

parameters. Input data included population-based prevalence surveys, case notifications from countries with high-quality cause of death data, estimates of excess mortality and remission, and population-based tuberculin surveys.

Consistent with previous GBD cycles, we harmonised prevalence surveys that used smear-positive rather than bacteriologically confirmed TB definitions and recalibrated surveys relying on symptoms-only screening compared with those using both symptoms and chest radiography. Because prevalence surveys primarily capture pulmonary TB, we further adjusted these data to account for extrapulmonary TB using notification data. For countries with sparse or low-quality cause-of-death data, we estimated incidence through a mortality-to-incidence ratio approach. Mortality-to-incidence ratios were modelled with a Bayesian meta-regression of locations with high-quality mortality data,²⁷ incorporating the HAQ Index²⁸ and GBD super-region fixed effects as covariates. These predicted ratios were then combined with estimated deaths to derive incidence for countries without high-quality cause-of-death data.

To improve coherence across parameters, priors for TB duration were informed by a systematic review of pre-chemotherapy studies and calibrated against the HAQ Index.²⁹ Final duration estimates were derived within DisMod-MR through statistical triangulation with observed incidence, prevalence, and mortality. As in earlier GBD iterations, we applied the estimated fraction of TB cases with HIV co-infection, modelled with data from the WHO TB notifications registrar as input (appendix 1 p 25), to distinguish TB morbidity between individuals living with HIV and those without HIV. We further disaggregated cases by drug-resistance status by applying predicted proportions of MDR-TB to all-form TB cases and to TB cases with HIV, consistent with the approach used for mortality. We further distinguished XDR-TB by estimating the proportion of MDR-TB cases that were XDR-TB at the super-region level and applying these proportions to MDR-TB estimates. MDR-TB and XDR-TB incidence includes both new and relapsed TB cases, as well as new cases arising among re-treated individuals (excluding relapse), thereby capturing both primary and acquired resistance (appendix 1 pp 25–27).

Risk factor analysis

Methods for estimating TB risk factor attribution have been described in detail previously.³⁰ In brief, we quantified PAFs for adults (aged ≥ 15 years) within the comparative risk assessment framework of GBD 2023. Inputs for the PAF calculation included age-sex-specific prevalence estimates for each risk factor; relative risks of TB mortality associated with each risk factor; and the theoretical minimum risk exposure level (TMREL), defined as the exposure level that minimises risk for each individual in the population.

For each risk–outcome pair, the number of deaths attributable to a risk factor was computed by multiplying the PAF by the total number of TB deaths among people without HIV. Risk-deleted mortality rates were subsequently derived to estimate the hypothetical TB mortality that would have been observed in 2023 if the risk factor had been set to its TMREL. This approach allows assessment of the potential impact of addressing key TB risk factors across locations, age groups, and sexes.

In this analysis, we included alcohol use,³¹ smoking,³² and high fasting plasma glucose as risk factors for tuberculosis among adults without HIV co-infection. Complete details of the methodology, including data sources, relative risk estimates, and uncertainty propagation, are provided in appendix 1 (pp 31–73).

Data presentation

TB estimates were computed at global, regional, and national levels with standard GBD regions comprising countries and territories that are geographically close and epidemiologically similar. Cause-of-death patterns were used to create the seven GBD super-regions: central Europe, eastern Europe, and central Asia; the high-income region; Latin America and the Caribbean; North Africa and the Middle East; south Asia; southeast Asia, east Asia, and Oceania; and sub-Saharan Africa (appendix 1 p 4).³³ We report GBD 2023 TB estimates as counts, all-age rates per 100 000 population, and age-standardised rates per 100 000 population using the GBD standard population. Disability-adjusted life-years (DALYs) were calculated as the sum of years of life lost (YLLs) and years lived with disability (YLDs). To assess interim WHO End TB milestone progress, we present percentage changes in TB burden in 2023 relative to 2015 for both rates and counts. Following GBD rounding standard guidance, counts are rounded to three significant figures, while all rates, percentages (including percentage changes), and absolute differences are rounded to one decimal place. Throughout the remainder of the analysis, MDR-TB estimates include XDR-TB (unless otherwise specified), as XDR-TB is a subset of MDR-TB. Drug-susceptible TB includes all TB cases not meeting the criteria for MDR-TB and therefore encompasses non-MDR drug-resistant forms, such as monoresistant TB forms. Parameter uncertainty was incorporated at each modelling step by drawing 250 samples from the age-sex-location-year-specific parameter distributions and propagating these through all subsequent calculations. 95% uncertainty intervals (UIs) were derived from the 2.5th and 97.5th percentiles of the final 250 draws. More detailed TB burden results by HIV status, age, sex, location, year, and further drug-resistance disaggregation are available in the GBD Results Tool.

Role of the funding source

The funder of the study had no role in the study design, data collection, data analysis, data interpretation, writing

of the report, or the decision to submit the manuscript for publication.

Results

Overview

In 2023, an estimated 9.11 million (95% UI 8.04–10.3) incident cases of all-form TB and 1.22 million (0.98–1.49) deaths occurred globally (table 1), corresponding to an all-age incidence rate of 112.9 (99.6–127.5) per 100 000 population and a mortality rate of 15.1 (12.1–18.4) per 100 000. Of all TB cases, 781 000 (690 000–879 000) incident cases and 210 000 (142 000–279 000) deaths were HIV-related TB (appendix 2 table S1). All-form TB resulted in an estimated 54.6 million (43.8–65.5) DALYs worldwide in 2023, comprising 48.0 million (37.4–58.5) YLLs and 6.62 million (4.53–9.02) YLDs (table 1). HIV-related TB accounted for an estimated 11.0 million (7.56–14.3) DALYs, including 10.3 million (6.94–13.7) YLLs and 643 000 (433 000–853 000) YLDs (appendix 2 table S1).

The incidence of HIV-related TB in 2023 was concentrated in sub-Saharan Africa, with an estimated 603 000 (526 000–678 000) cases (appendix 2 table S1). Conversely, the number of non-HIV-related TB incident cases was the highest in south Asia, at 2.94 million (2.56–3.35), followed by sub-Saharan Africa, at 2.61 million (2.24–2.99), and southeast Asia, east Asia, and Oceania, at 2.05 million (1.84–2.25; appendix 2 table S1). India had the largest number of non-HIV-related TB incident cases, estimated at 2.22 million (1.92–2.54), followed by China (652 000 [582 000–723 000]), Indonesia (455 000 [401 000–506 000]), the Philippines (411 000 [365 000–450 000]), and Pakistan (410 000 [357 000–471 000]).

The age-standardised mortality rate for both HIV-related and non-HIV-related TB in 2023 was highest in sub-Saharan Africa, estimated at 20.2 (13.5–27.3) per 100 000 population and 50.3 (33.9–69.6) per 100 000 population, respectively (appendix 2 table S1). Lesotho had the highest age-standardised HIV-related TB mortality rate, at 165.4 (133.4–203.6) per 100 000, followed by Eswatini at 149.5 (95.3–193.8) per 100 000 and Zimbabwe at 76.1 (51.1–101.6) per 100 000. The age-standardised non-HIV-related TB mortality rate was highest in the Central African Republic (240.6 [100.3–464.5] per 100 000), followed by Somalia (128.1 [65.4–201.5] per 100 000) and Lesotho (116.0 [70.2–183.5] per 100 000).

In the past two decades, the global age-standardised mortality rate for HIV-related TB changed by –33.1% (–39.6 to –23.1) between 2000 and 2010, and by a further –55.6% (–64.5 to –47.0) between 2010 and 2023 (appendix 2 figure S1; tables S7, S8). In comparison, the global age-standardised mortality rate for non-HIV-related TB changed by –33.9% (–44.7 to –19.9) between 2000 and 2010 and by –46.1% (–60.5 to –28.6) between 2010 and 2023 (appendix 2 figure S1; tables S7, S8).

Global drug-resistant TB burden

In 2023, MDR-TB, including XDR-TB, accounted for 466 000 (95% UI 198 000–1080 000) cases, and contributed to 102 000 (31700–238 000) deaths (table 1). MDR-TB resulted in 3.96 million (1.31–9.01) DALYs, with 3.68 million (1.18–8.31) YLLs and 277 000 (106 000–662 000) YLDs. In 2023, the highest percentages of drug-resistant TB (among people without HIV) relative to all-form TB incidence were observed in Moldova (47.6%), Belarus (45.0%), and Ukraine (35.1%). The

highest percentages of drug-resistant TB deaths (among people without HIV) relative to all-form TB mortality were observed in Moldova (58.6%), Belarus (56.3%), and Kyrgyzstan (41.6%; figure 1). Most of the MDR-TB burden arose from all-form MDR-TB without extensive resistance, equalling 469 000 (200 000–1090 000) incident cases (appendix 2 table S2). Among these, 442 000 incident cases (186 000–1030 000) occurred in individuals without HIV and 27 000 cases (14100–50 600) occurred in those living with HIV. By contrast, there were 24200

	DALYs		YLDs		YLLs		Incidence		Deaths	
	Count	Rate per 100 000	Count	Rate per 100 000	Count	Rate per 100 000	Count	Rate per 100 000	Count	Rate per 100 000
Global										
All-form TB	54 600 000 (43 800 000 to 65 500 000)	676.8 (543.0 to 812.5)	6 620 000 (4 530 000 to 9 020 000)	82.1 (56.2 to 111.9)	48 000 000 (37 400 000 to 58 500 000)	594.7 (464.2 to 724.8)	9 110 000 (8 040 000 to 10 300 000)	112.9 (99.6 to 127.5)	1 220 000 (975 000 to 1 490 000)	15.1 (12.1 to 18.4)
Drug-susceptible TB	39 700 000 (30 800 000 to 49 500 000)	491.7 (381.9 to 613.9)	5 700 000 (3 870 000 to 7 930 000)	70.7 (48.0 to 98.3)	34 000 000 (24 900 000 to 43 300 000)	420.9 (308.5 to 537.4)	7 860 000 (6 810 000 to 8 880 000)	97.5 (84.5 to 110.0)	908 000 (671 000 to 1 140 000)	11.3 (8.3 to 14.1)
MDR-TB	3 960 000 (1 310 000 to 9 010 000)	49.1 (16.2 to 111.8)	277 000 (106 000 to 662 000)	3.4 (1.3 to 8.2)	3 680 000 (1 180 000 to 8 310 000)	45.6 (14.7 to 103.1)	466 000 (198 000 to 1 080 000)	5.8 (2.5 to 13.4)	102 000 (31 700 to 238 000)	1.3 (0.4 to 2.9)
HIV-related drug-susceptible TB	9 980 000 (6 850 000 to 13 100 000)	123.7 (84.9 to 162.1)	614 000 (415 000 to 826 000)	7.6 (5.1 to 10.2)	9 360 000 (6 140 000 to 12 500 000)	116.1 (76.1 to 154.5)	753 000 (659 000 to 852 000)	9.3 (8.2 to 10.6)	190 000 (125 000 to 253 000)	2.4 (1.6 to 3.1)
HIV-related MDR-TB	995 000 (356 000 to 2 040 000)	12.3 (4.4 to 25.3)	28 700 (12 200 to 57 800)	0.4 (0.2 to 0.7)	966 000 (336 000 to 2 000 000)	12.0 (4.2 to 24.8)	28 100 (14 700 to 52 400)	0.3 (0.2 to 0.7)	19 600 (6840 to 40 500)	0.2 (0.1 to 0.5)
Central Europe, eastern Europe, and central Asia										
All-form TB	643 000 (592 000 to 700 000)	152.0 (139.9 to 165.5)	65 600 (43 600 to 91 600)	15.5 (10.3 to 21.6)	578 000 (535 000 to 628 000)	136.5 (126.5 to 148.5)	212 000 (177 000 to 255 000)	50.1 (41.8 to 60.4)	14 800 (13 900 to 16 000)	3.5 (3.3 to 3.8)
Drug-susceptible TB	356 000 (231 000 to 458 000)	84.1 (54.7 to 108.3)	46 900 (29 300 to 67 100)	11.1 (6.9 to 15.9)	309 000 (199 000 to 406 000)	73.0 (46.9 to 96.0)	141 000 (105 000 to 178 000)	33.4 (24.8 to 42.1)	8230 (5410 to 10 800)	1.9 (1.3 to 2.6)
MDR-TB	211 000 (108 000 to 329 000)	49.8 (25.6 to 77.7)	15 200 (7840 to 27 200)	3.6 (1.9 to 6.4)	196 000 (101 000 to 306 000)	46.2 (23.8 to 72.4)	62 600 (32 700 to 115 000)	14.8 (7.7 to 27.2)	5100 (2690 to 7940)	1.2 (0.6 to 1.9)
HIV-related drug-susceptible TB	38 200 (16 800 to 65 300)	9.0 (4.0 to 15.4)	2320 (1240 to 3540)	0.5 (0.3 to 0.8)	35 900 (15 000 to 62 300)	8.5 (3.6 to 14.7)	6060 (3450 to 8400)	1.4 (0.8 to 2.0)	729 (305 to 1260)	0.2 (0.1 to 0.3)
HIV-related MDR-TB	38 500 (15 300 to 70 500)	9.1 (3.6 to 16.7)	1210 (565 to 2290)	0.3 (0.1 to 0.5)	37 300 (14 400 to 68 900)	8.8 (3.4 to 16.3)	2150 (978 to 4110)	0.5 (0.2 to 1.0)	760 (294 to 1400)	0.2 (0.1 to 0.3)
High income										
All-form TB	302 000 (267 000 to 333 000)	27.3 (24.2 to 30.2)	35 600 (23 700 to 48 900)	3.2 (2.2 to 4.4)	266 000 (235 000 to 295 000)	24.1 (21.3 to 26.8)	90 100 (78 700 to 103 000)	8.2 (7.1 to 9.4)	13 100 (11 200 to 14 600)	1.2 (1.0 to 1.3)
Drug-susceptible TB	228 000 (203 000 to 251 000)	20.6 (18.4 to 22.8)	31 500 (20 900 to 43 100)	2.9 (1.9 to 3.9)	196 000 (172 000 to 218 000)	17.8 (15.6 to 19.7)	82 400 (71 900 to 94 900)	7.5 (6.5 to 8.6)	11 300 (9380 to 12 800)	1.0 (0.9 to 1.2)
MDR-TB	7610 (3310 to 16 600)	0.7 (0.3 to 1.5)	548 (294 to 1080)	0.0 (0.0 to 0.1)	7060 (2890 to 15 800)	0.6 (0.3 to 1.4)	1570 (996 to 2890)	0.1 (0.1 to 0.3)	390 (166 to 856)	0.0 (0.0 to 0.1)
HIV-related drug-susceptible TB	63 400 (44 000 to 80 800)	5.7 (4.0 to 7.3)	3500 (2200 to 4910)	0.3 (0.2 to 0.4)	59 900 (39 700 to 77 900)	5.4 (3.6 to 7.1)	6010 (5200 to 6950)	0.5 (0.5 to 0.6)	1420 (936 to 1850)	0.1 (0.1 to 0.2)
HIV-related MDR-TB	2760 (852 to 6570)	0.3 (0.1 to 0.6)	67 (31 to 157)	0.0 (0.0 to 0.0)	2700 (809 to 6470)	0.2 (0.1 to 0.6)	87 (45 to 175)	0.0 (0.0 to 0.0)	64 (20 to 152)	0.0 (0.0 to 0.0)

(Table 1 continues on next page)

	DALYs		YLDs		YLLs		Incidence		Deaths	
	Count	Rate per 100 000	Count	Rate per 100 000	Count	Rate per 100 000	Count	Rate per 100 000	Count	Rate per 100 000
(Continued from previous page)										
Latin America and Caribbean										
All-form TB	1 660 000 (1 400 000 to 1 920 000)	280.1 (236.2 to 323.7)	165 000 (109 000 to 231 000)	27.9 (18.4 to 39.1)	1 490 000 (1 250 000 to 1 760 000)	252.2 (210.3 to 297.3)	245 000 (208 000 to 285 000)	41.3 (35.2 to 48.1)	36 000 (30 900 to 41 200)	6.1 (5.2 to 7.0)
Drug-susceptible TB	1 120 000 (948 000 to 1 320 000)	189.7 (160.1 to 222.6)	136 000 (90 600 to 192 000)	22.9 (15.3 to 32.4)	988 000 (812 000 to 1 160 000)	166.8 (137.1 to 196.3)	206 000 (176 000 to 239 000)	34.7 (29.7 to 40.3)	25 400 (21 200 to 29 600)	4.3 (3.6 to 5.0)
MDR-TB	72 600 (30 400 to 169 000)	12.3 (5.1 to 28.6)	4650 (1970 to 11 000)	0.8 (0.3 to 1.9)	67 900 (27 600 to 162 000)	11.5 (4.7 to 27.3)	8360 (3860 to 20 600)	1.4 (0.7 to 3.5)	1860 (750 to 4480)	0.3 (0.1 to 0.8)
HIV-related drug-susceptible TB	435 000 (289 000 to 589 000)	73.5 (48.8 to 99.4)	24 000 (15 900 to 32 400)	4.1 (2.7 to 5.5)	411 000 (268 000 to 565 000)	69.5 (45.2 to 95.4)	30 100 (25 600 to 34 900)	5.1 (4.3 to 5.9)	8130 (5390 to 11 200)	1.4 (0.9 to 1.9)
HIV-related MDR-TB	27 400 (7470 to 74 900)	4.6 (1.3 to 12.6)	822 (280 to 2310)	0.1 (0.0 to 0.4)	26 600 (7020 to 72 900)	4.5 (1.2 to 12.3)	764 (270 to 2110)	0.1 (0.0 to 0.4)	558 (146 to 1550)	0.1 (0.0 to 0.3)
North Africa and Middle East										
All-form TB	788 000 (539 000 to 1 140 000)	123.5 (84.5 to 178.0)	148 000 (98 100 to 203 000)	23.2 (15.4 to 31.8)	640 000 (393 000 to 993 000)	100.4 (61.5 to 155.6)	226 000 (193 000 to 260 000)	35.4 (30.2 to 40.8)	16 800 (11 200 to 26 600)	2.6 (1.7 to 4.2)
Drug-susceptible TB	692 000 (476 000 to 992 000)	108.4 (74.6 to 155.5)	138 000 (92 600 to 192 000)	21.7 (14.5 to 30.1)	554 000 (343 000 to 863 000)	86.7 (53.7 to 135.3)	214 000 (182 000 to 247 000)	33.5 (28.5 to 38.7)	14 800 (9840 to 23 700)	2.3 (1.5 to 3.7)
MDR-TB	43 400 (14 300 to 105 000)	6.8 (2.2 to 16.5)	4320 (2060 to 9610)	0.7 (0.3 to 1.5)	39 100 (11 900 to 98 600)	6.1 (1.9 to 15.4)	6110 (3140 to 13 900)	1.0 (0.5 to 2.2)	1000 (311 to 2490)	0.2 (0.0 to 0.4)
HIV-related drug-susceptible TB	49 900 (28 800 to 82 900)	7.8 (4.5 to 13.0)	4940 (3220 to 6960)	0.8 (0.5 to 1.1)	44 900 (23 900 to 79 100)	7.0 (3.7 to 12.4)	6010 (5200 to 6890)	0.9 (0.8 to 1.1)	890 (474 to 1560)	0.1 (0.1 to 0.2)
HIV-related MDR-TB	2920 (854 to 7990)	0.5 (0.1 to 1.3)	133 (61 to 301)	0.0 (0.0 to 0.0)	2790 (746 to 7810)	0.4 (0.1 to 1.2)	89 (41 to 201)	0.0 (0.0 to 0.0)	55 (15 to 155)	0.0 (0.0 to 0.0)
South Asia										
All-form TB	16 400 000 (13 400 000 to 20 600 000)	868.1 (706.2 to 1089.4)	2 180 000 (1 520 000 to 2 980 000)	115.4 (80.2 to 157.5)	14 200 000 (11 200 000 to 18 200 000)	752.7 (593.1 to 963.8)	3 000 000 (2 610 000 to 3 410 000)	158.3 (138.0 to 180.2)	426 000 (336 000 to 550 000)	22.5 (17.8 to 29.1)
Drug-susceptible TB	13 600 000 (8 990 000 to 17 800 000)	719.4 (475.2 to 938.5)	1 980 000 (1 340 000 to 2 770 000)	104.7 (70.9 to 146.4)	11 600 000 (7 440 000 to 15 800 000)	614.7 (393.1 to 833.8)	2 710 000 (2 240 000 to 3 160 000)	143.4 (118.3 to 166.7)	353 000 (222 000 to 484 000)	18.7 (11.7 to 25.6)
MDR-TB	2 110 000 (311 000 to 5 560 000)	111.4 (16.4 to 293.6)	151 000 (22 600 to 400 000)	8.0 (1.2 to 21.1)	1 960 000 (284 000 to 5 200 000)	103.4 (15.0 to 274.7)	226 000 (35 700 to 599 000)	11.9 (1.9 to 31.6)	59 700 (8870 to 162 000)	3.2 (0.5 to 8.5)
HIV-related drug-susceptible TB	595 000 (357 000 to 848 000)	31.4 (18.9 to 44.8)	47 500 (31 400 to 66 800)	2.5 (1.7 to 3.5)	547 000 (310 000 to 799 000)	28.9 (16.4 to 42.2)	52 200 (40 200 to 61 300)	2.8 (2.1 to 3.2)	11 400 (6440 to 16 400)	0.6 (0.3 to 0.9)
HIV-related MDR-TB	112 000 (15 600 to 332 000)	5.9 (0.8 to 17.5)	4530 (601 to 13 000)	0.2 (0.0 to 0.7)	107 000 (14 600 to 316 000)	5.7 (0.8 to 16.7)	3260 (491 to 9410)	0.2 (0.0 to 0.5)	2230 (309 to 6550)	0.1 (0.0 to 0.3)

(Table 1 continues on next page)

(12 300–45 300) incident cases of XDR-TB among individuals without HIV and 1130 (614–1900) cases among people living with HIV (appendix 2 table S2).

Across GBD super-regions, the highest age-standardised incidence rates of MDR-TB in 2023 were observed in central Europe, eastern Europe, and central Asia (13.1 [95% UI 6.8–23.8] per 100 000), south Asia (12.6 [2.0–33.0] per 100 000), and sub-Saharan Africa

(9.5 [4.5–21.0] per 100 000; appendix 2 table S3). The highest age-standardised mortality rates due to MDR-TB in 2023 were observed in south Asia (3.8 [0.6–10.3] per 100 000), followed by sub-Saharan Africa (3.5 [1.2–8.9] per 100 000). The age-standardised mortality rate of MDR-TB in central Europe, eastern Europe, and central Asia was estimated at 0.9 (0.5–1.5) per 100 000 (appendix 2 table S3).

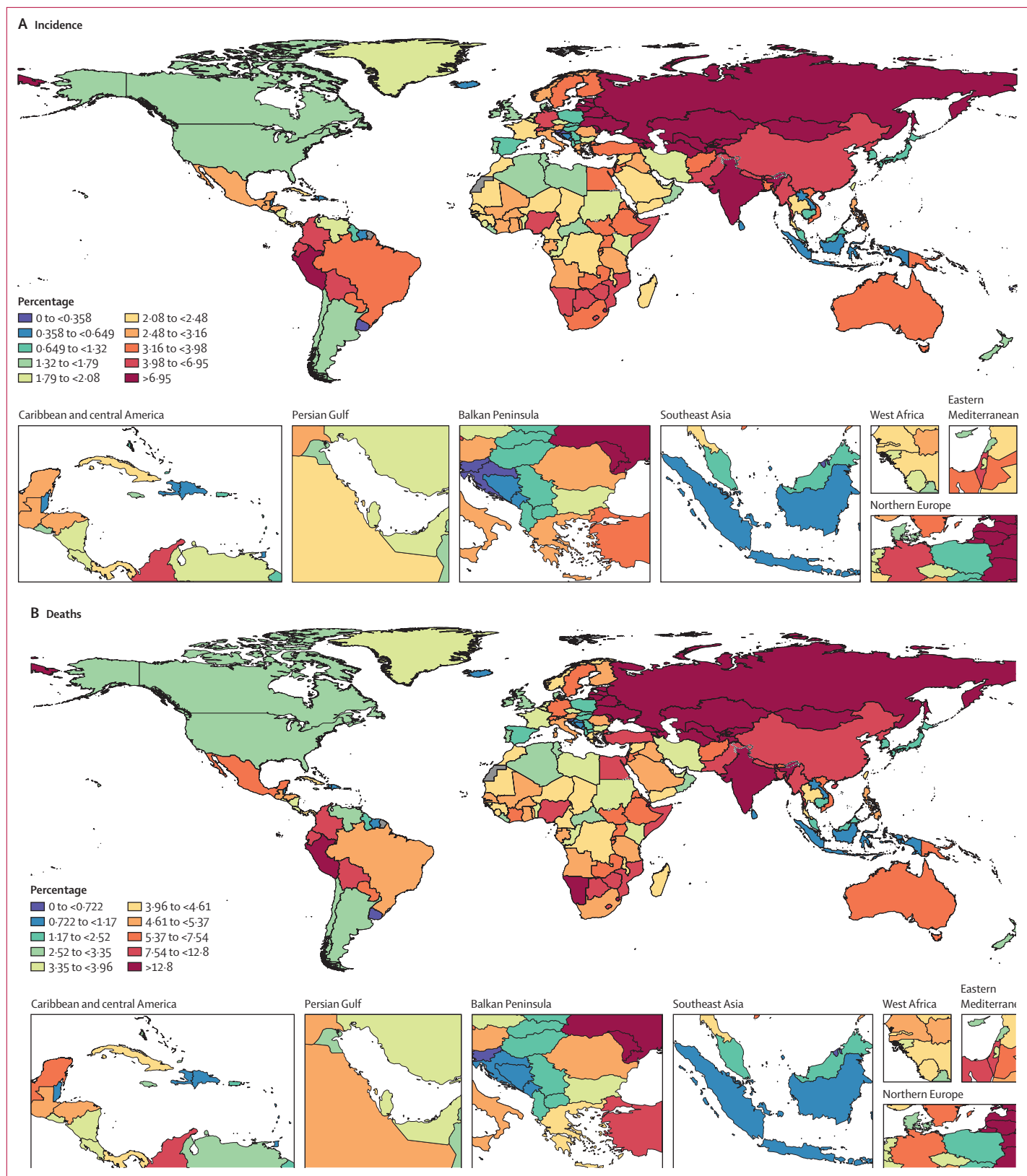
	DALYs		YLDs		YLLs		Incidence		Deaths	
	Count	Rate per 100 000	Count	Rate per 100 000	Count	Rate per 100 000	Count	Rate per 100 000	Count	Rate per 100 000
(Continued from previous page)										
Southeast Asia, east Asia, and Oceania										
All-form TB	8 180 000 (6 840 000 to 9 500 000)	370.1 (309.6 to 430.0)	1 530 000 (1 020 000 to 2 030 000)	69.2 (46.3 to 91.9)	6 650 000 (5 240 000 to 7 960 000)	300.9 (237.2 to 360.1)	2 120 000 (1 910 000 to 2 330 000)	96.1 (86.4 to 105.5)	195 000 (155 000 to 235 000)	8.8 (7.0 to 10.6)
Drug-susceptible TB	7 160 000 (5 800 000 to 8 390 000)	323.9 (262.6 to 379.9)	1 420 000 (953 000 to 1 910 000)	64.3 (43.1 to 86.4)	5 730 000 (4 580 000 to 7 000 000)	259.6 (207.5 to 316.9)	1 990 000 (1 760 000 to 2 200 000)	89.9 (79.9 to 99.4)	172 000 (138 000 to 209 000)	7.8 (6.3 to 9.5)
MDR-TB	347 000 (137 000 to 728 000)	15.7 (6.2 to 33.0)	39 000 (16 700 to 89 500)	1.8 (0.8 to 4.1)	308 000 (114 000 to 679 000)	13.9 (5.2 to 30.8)	65 200 (27 200 to 167 000)	2.9 (1.2 to 7.6)	9810 (3630 to 21 200)	0.4 (0.2 to 1.0)
HIV-related drug-susceptible TB	619 000 (427 000 to 793 000)	28.0 (19.3 to 35.9)	65 800 (44 100 to 91 000)	3.0 (2.0 to 4.1)	553 000 (367 000 to 739 000)	25.0 (16.6 to 33.5)	69 700 (62 000 to 78 100)	3.2 (2.8 to 3.5)	11 600 (7690 to 15 600)	0.5 (0.3 to 0.7)
HIV-related MDR-TB	54 700 (21 100 to 107 000)	2.5 (1.0 to 4.8)	2950 (1450 to 5640)	0.1 (0.1 to 0.3)	51 800 (18 800 to 103 000)	2.3 (0.8 to 4.7)	1850 (1010 to 3600)	0.1 (0.0 to 0.2)	1080 (397 to 2090)	0.0 (0.0 to 0.1)
Sub-Saharan Africa										
All-form TB	26 600 000 (19 100 000 to 33 800 000)	2203.2 (1584.9 to 2803.3)	2 500 000 (1 670 000 to 3 450 000)	206.9 (138.4 to 285.8)	24 100 000 (16 600 000 to 31 400 000)	1996.3 (1372.1 to 2601.1)	3 220 000 (2 780 000 to 3 650 000)	266.5 (230.0 to 302.2)	518 000 (362 000 to 670 000)	43.0 (30.0 to 55.5)
Drug-susceptible TB	16 500 000 (11 100 000 to 22 700 000)	1365.9 (920.2 to 1877.4)	1 950 000 (1 300 000 to 2 760 000)	161.6 (108.0 to 228.4)	14 500 000 (9 340 000 to 20 900 000)	1204.3 (774.2 to 1729.1)	2 520 000 (2 160 000 to 2 890 000)	208.6 (178.7 to 239.4)	323 000 (212 000 to 461 000)	26.7 (17.6 to 38.2)
MDR-TB	1 170 000 (376 000 to 2 840 000)	96.8 (31.1 to 235.3)	62 100 (24 000 to 145 000)	5.1 (2.0 to 12.1)	1 110 000 (339 000 to 2 730 000)	91.7 (28.1 to 226.3)	96 400 (45 300 to 219 000)	8.0 (3.8 to 18.1)	24 400 (7760 to 61 300)	2.0 (0.6 to 5.1)
HIV-related drug-susceptible TB	8 180 000 (5 500 000 to 10 800 000)	677.8 (455.7 to 896.8)	466 000 (317 000 to 630 000)	38.6 (26.3 to 52.2)	7 710 000 (4 950 000 to 10 300 000)	639.2 (410.0 to 849.7)	583 000 (510 000 to 656 000)	48.3 (42.2 to 54.4)	156 000 (101 000 to 208 000)	13.0 (8.4 to 17.2)
HIV-related MDR-TB	757 000 (257 000 to 1 650 000)	62.7 (21.3 to 136.8)	19 000 (8060 to 40 800)	1.6 (0.7 to 3.4)	738 000 (247 000 to 1 620 000)	61.2 (20.5 to 133.9)	19 900 (10 300 to 39 900)	1.6 (0.9 to 3.3)	14 800 (5060 to 31 800)	1.2 (0.4 to 2.6)
Data in parentheses are 95% uncertainty intervals. All-form TB represents the sum of four subsequent rows (drug-susceptible TB, MDR-TB, HIV-related drug-susceptible TB, and HIV-related MDR-TB). MDR-TB estimates include XDR-TB. For clarification, drug-susceptible TB and MDR-TB refer to TB among people without HIV. To be consistent with the GBD classification, drug-susceptible TB includes all TB cases not meeting the criteria for MDR-TB. While primarily defined as being susceptible to isoniazid and rifampicin, this reporting category also encompasses other non-MDR drug-resistant strains, such as isoniazid-monoresistant TB. DALY=disability-adjusted life-year. GBD=Global Burden of Diseases, Injuries, and Risk Factors Study. MDR=multidrug-resistant. TB=tuberculosis. YLD=years lived with disability. YLL=years of life lost. XDR=extensively drug resistant.										
Table 1: Tuberculosis DALYs, YLDs, YLLs, incidence, and deaths with corresponding all-age rates by drug resistance, HIV status, and GBD super-region in 2023										

Globally, the age-standardised incidence rate of MDR-TB increased substantially between 1990 and 2000 (GBD Results Tool), and improvements were observed between 2000 and 2023. Between 2000 and 2010, age-standardised incidence rates of MDR-TB changed by -9.3% (-59.0 to 119.9). A greater change of -27.8% (-71.8 to 46.1) was observed between 2010 and 2023 (figure 2; appendix 2 tables S5, S6). Considerable geographical variation was observed in these temporal patterns. Between 1990 and 2000, MDR-TB incidence increased the most in south Asia; central Europe, eastern Europe, and central Asia; and sub-Saharan Africa (GBD Results Tool). In the following decade (2000–10), considerable reductions were observed in southeast Asia, east Asia, and Oceania (-57.6% [-76.8 to -22.5]) and in the high-income super-region (-46.1% [-62.7 to -18.3]), while the direction of change in other super-regions was

largely uncertain. From 2010 to 2023, no substantial changes were detected in any super-region (95% UIs for all super-regions crossed zero).

The age-standardised mortality rate of MDR-TB showed a similar temporal pattern, increasing substantially globally and across all super-regions between 1990 and 2000, followed by heterogeneous change in subsequent decades (GBD Results Tool; figure 2). Marked declines in MDR-TB mortality between 2000 and 2010 were observed in the high-income super-region, with a change in the age-standardised mortality rate of -59.0% (-71.9 to -34.7), with a further change between 2010 and 2023 of -39.3% (-63.0 to -2.1). In southeast Asia, east Asia, and Oceania, the age-standardised mortality rate changed by -60.6% (-78.4 to -32.3) between 2000 and 2010 and by a further -61.0% (-81.3 to -16.8) between 2010 and 2023.

For the GBD Results Tool see
<https://vizhub.healthdata.org/gbd-results/>



(Figure 1 continues on next page)

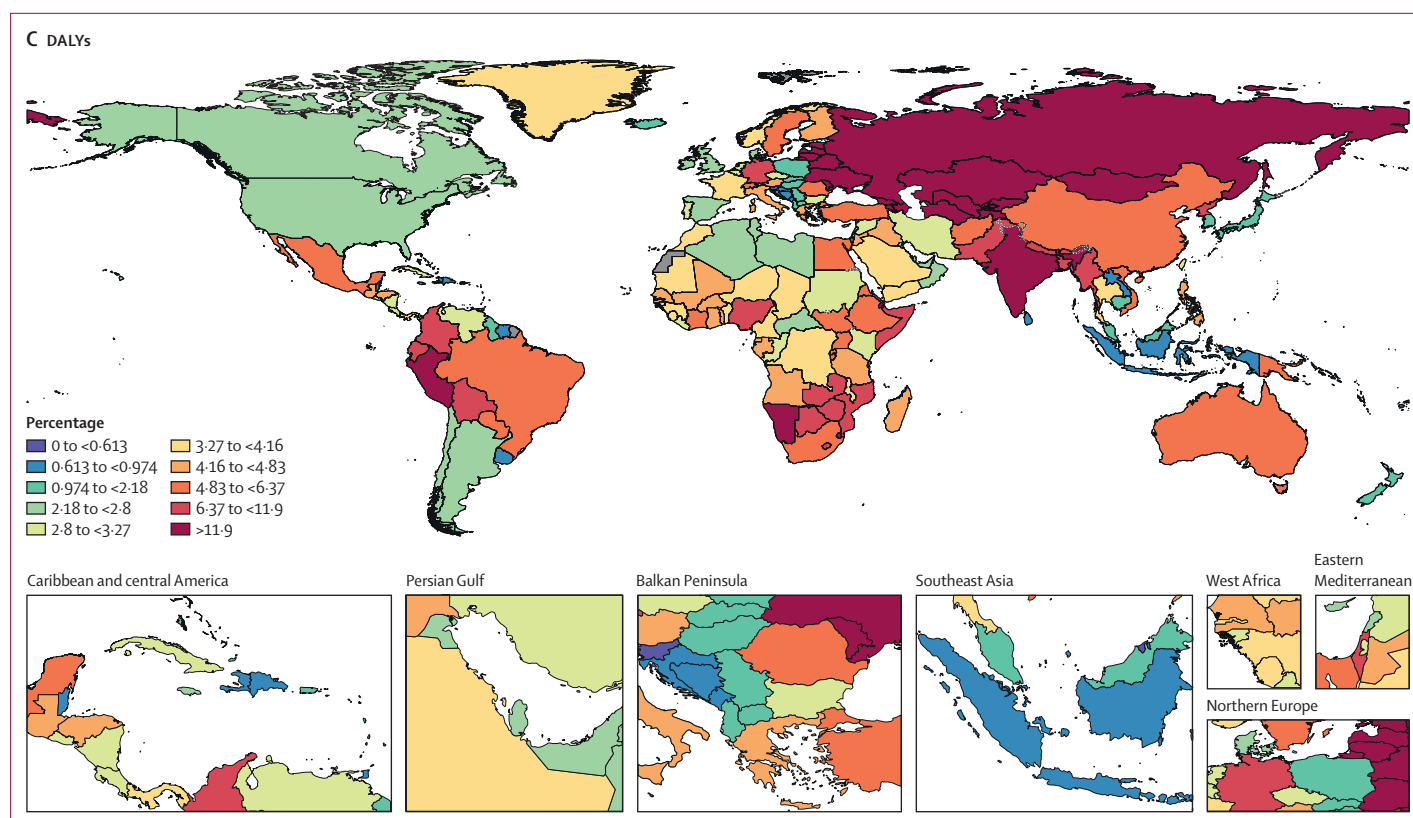


Figure 1: Proportion of all-form tuberculosis incidence (A), deaths (B), and DALYs (C) caused by multidrug-resistant tuberculosis in 2023
DALY=disability-adjusted life-year.

In central Europe, eastern Europe, and central Asia, changes were evident between 2010 and 2023, with an estimated change of -67.1% (-83.8 to -29.4 ; appendix 2 table S6). In other super-regions, the estimated changes were highly uncertain across both periods.

Risk-deleted TB burden

In 2023, the largest share of the TB burden was attributable to smoking, contributing to 12.3% (95% UI 9.2 – 15.8) of deaths, 10.6% (7.8 – 13.7) of YLLs, and 10.4% (7.8 – 13.3) of DALYs (appendix 2 table S4). Alcohol use contributed to 6.6% (1.0 – 13.2) of TB deaths, 6.7% (1.0 – 13.4) of YLLs, and 6.7% (1.0 – 13.4) of DALYs, while high fasting plasma glucose was associated with 8.0% (5.3 – 11.1) of deaths, 5.3% (3.6 – 7.4) of YLLs, and 5.2% (3.4 – 7.3) of DALYs (appendix 2 table S4). After removing the combined effects of these three major risk factors for TB, the estimated number of global deaths from all-form TB in 2023 would decrease from 1.22 million (0.98 – 1.49) to 768 000 ($592\,000$ – $970\,000$; tables 1, 2). YLLs would decline from 48.0 million (37.4 – 58.5) to 30.0 million (23.3 – 38.3) and DALYs would decline from 54.6 million (43.8 – 65.5) to 34.9 million (27.8 – 43.8). For MDR-TB, global deaths would decrease from 102 000 ($31\,700$ – $238\,000$) to 77 200 ($23\,400$ – $183\,000$) and global

DALYs would decrease from 3.96 million (1.31 – 9.01) to 3.12 million (1.03 – 7.29) after removal of these three major tuberculosis risk factors (tables 1, 2).

End TB milestones

At the global level, the all-age incidence rate of all-form TB changed by -19.2% (95% UI -20.5 to -17.8) between 2015 and 2023, from 139.8 cases (122.4 to 157.2) per 100 000 to 112.9 (99.6 to 127.5) per 100 000 (table 3). When stratified by drug-resistance status, the global incidence rate of drug-susceptible TB changed by -17.9% (-21.2 to -15.5), while the incidence rate for MDR-TB changed by -11.4% (-48.7 to 41.2 ; table 3). Among GBD super-regions, the largest reductions in all-form TB incidence were observed in central Europe, eastern Europe, and central Asia (-30.5% [-36.6 to -22.7]), sub-Saharan Africa (-25.8% [-27.3 to -24.1]), and south Asia (-25.7% [-28.0 to -23.1]; table 3). The smallest reductions occurred in southeast Asia, east Asia, and Oceania (-12.7% [-14.0 to -11.1]) and north Africa and the Middle East (-15.4% [-17.7 to -12.9]; table 3). In Latin America and the Caribbean, the all-form TB incidence rate changed by 0.6% (-2.2 to 3.4) between 2015 and 2023 (table 3). The magnitude of the decline in incidence was greater for drug-susceptible

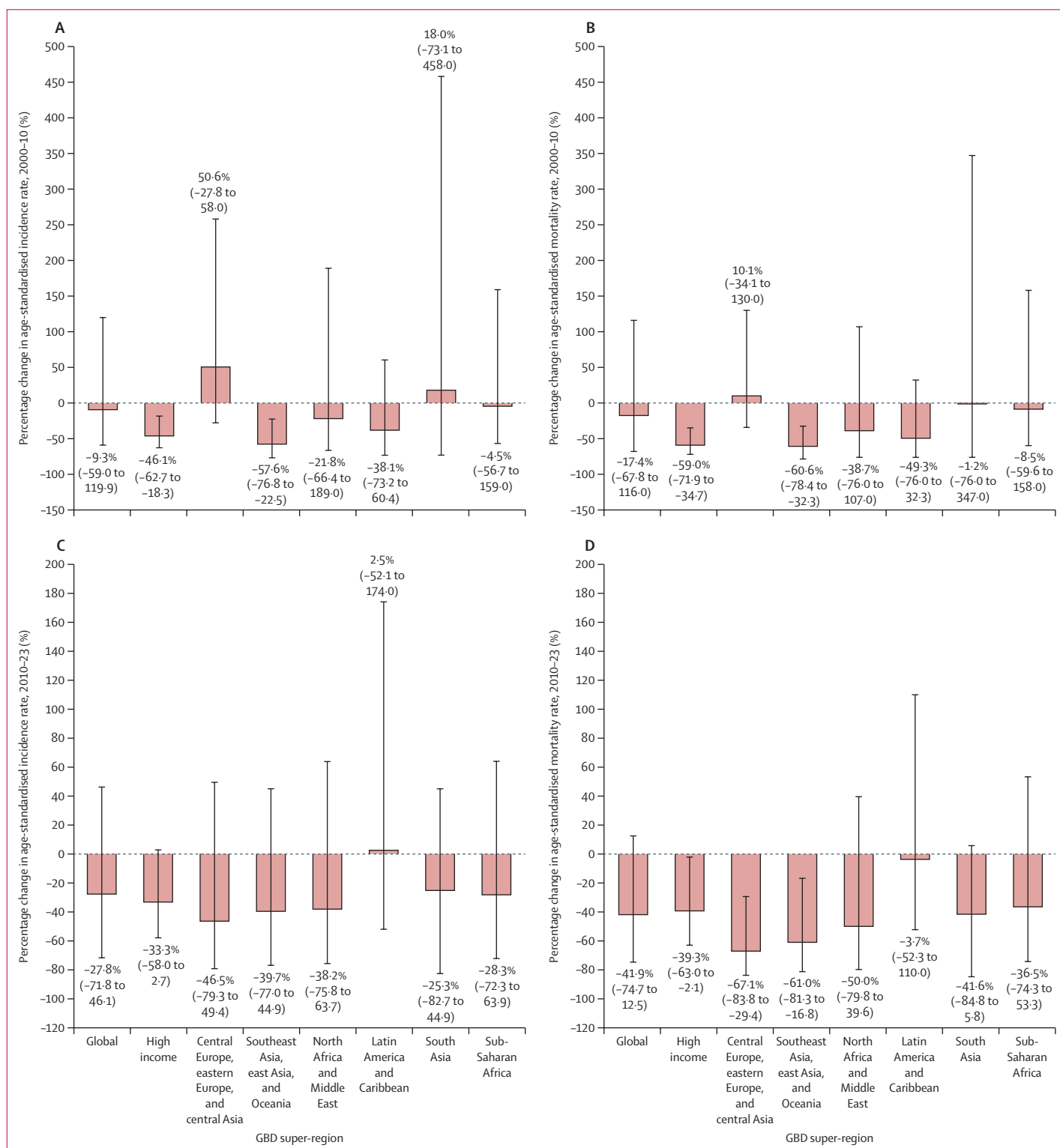


Figure 2: Percentage change in age-standardised incidence and mortality rates of multidrug-resistant tuberculosis in 2000–10 and 2010–23, globally and by GBD super-region. Error bars represent 95% uncertainty intervals. GBD=Global Burden of Diseases, Injuries, and Risk Factors Study.

TB than for MDR-TB across most regions. Changes in MDR-TB incidence rates were inconclusive (table 3).

Globally, the number of deaths due to all-form TB changed by -22.6% (95% UI -35.7 to -4.7), from 1.58 million (1.34–1.94) in 2015 to 1.22 million (0.975–1.49) in 2023 (table 3). Stratified by drug-resistance status, deaths from drug-susceptible TB changed by -21.7% (-36.7 to 0.1), and deaths from MDR-TB changed by -13.6% (-49.7 to 33.0), with considerable uncertainty (table 3). GBD super-regions with noticeable reductions in all-form TB deaths between 2015 and 2023 included central Europe, eastern Europe, and central Asia (-44.9% [-48.8 to -41.2]), south Asia (-27.7% [-42.0 to -6.2]), and southeast Asia, east Asia, and Oceania (-22.4% [-35.3 to -4.9]; table 3). Estimates in other super-regions were highly uncertain.

Discussion

This study presents the latest results, from GBD 2023, on the global burden of TB in the context of HIV and multidrug resistance, highlighting the precariousness of gains achieved and the complexity of TB containment. With more than 9 million incident cases and 1.22 million deaths, TB was the leading cause of death by a single infectious pathogen in 2023. Among all TB cases, an estimated 781 000 cases were linked to HIV infection, with 210 000 deaths—corresponding to more than one in six of all TB fatalities. MDR-TB represented approximately 5% of all-form TB incidence in 2023 but accounted for 7% of global all-form TB DALYs, illustrating its disproportionate severity. The year 2015 was a watershed point for the global TB response, marking the transition from the UN Millennium Development Goals to the UN Sustainable Development Goals, and from the WHO Stop TB Strategy to the WHO End TB Strategy.³⁴ Benchmarking against 2015, the global TB incidence rate fell by about 19% and deaths fell by 23%. However, these gains remain insufficient to meet the WHO End TB Strategy milestones of a 50% reduction in incidence and a 75% reduction in deaths by 2025, and remain far from the 2035 targets of 90% and 95% reductions, respectively. Improvements have been uneven across geographical regions. While central Europe, eastern Europe, and central Asia have seen some of the steepest declines in both incidence and mortality, progress has stalled in Latin America and the Caribbean. Taken together, these patterns suggest that progress towards TB elimination has been uneven, highlighting the need for intensified and locally tailored prevention, diagnostic, and treatment strategies in settings where declines in TB incidence and mortality have stagnated.

HIV remains a persistent driver of TB burden, exacerbating global disparities. Although HIV accounted for only 8.6% of all-form TB globally in 2023, it contributed disproportionately to 17% of TB mortality. However, meaningful gains have been achieved in the past two decades. The decline in HIV-related TB mortality

after 2010 closely mirrored the rapid rollout of antiretroviral therapy (ART). In 2023, deaths related to HIV had fallen to less than half of their 2004 peak, from 1.71 million to 833 000 deaths.³⁵ From 2010 to 2023, TB mortality among those with HIV fell by about 56%. These synchronous trends suggest that ART accelerates TB prevention in HIV-epidemic areas.^{36,37} Through effective HIV suppression and reduced immune compromise, the risk that latent *Mycobacterium tuberculosis* infection progresses to active TB is substantially reduced, which in turn supports more favourable outcomes if *M tuberculosis* infection occurs.³⁶ However, the interdependence of the two epidemics exposes a pressing vulnerability, given the heavy reliance of global HIV programmes on foreign aid.^{38,39} Global developmental assistance for health has entered a period of protracted downturn, with an estimated 22% (US\$11.2 billion) decrease between 2024 and 2025, followed by an expected further 6% (\$2.2 billion) decrease between 2025 and 2030.⁴⁰ The USA is historically the largest contributor (72.6%) to international HIV funds through the President's Emergency Plan for AIDS Relief (PEPFAR). The early impact of a PEPFAR suspension that began on Jan 20, 2025, and was expected to last for 90 days was reported in 27 low-income and middle-income countries (LMICs).^{41,42} Although a subsequent waiver to PEPFAR was granted on Feb 1, 2025, during the 90-day review period, the waiver only permitted funding of "life-saving HIV services", such as diagnostics and treatment, and excluded most preventive services.⁴³ The redesign of the US Global Health Strategy released in September, 2025, promised the full resumption of PEPFAR funding through the fiscal year 2026, contingent on bilateral relationships and multi-year agreements with key countries;⁴⁴ however, the US Consolidated Appropriations Act enacted on Feb 3, 2026,⁴⁵ stipulated a \$91.2 million budget reduction for PEPFAR and a \$400 million decrease in contributions to the Global Fund to Fight AIDS, Tuberculosis and Malaria.⁴⁶ Service disruption resulting from funding cuts threatens the lives of millions and risks reversing decades of progress in HIV control.³⁵ Because the TB burden among people living with HIV is tightly linked to ART coverage, this contraction could translate to substantial increases in TB incidence and mortality. A modelling study projected approximately 1.4 million additional TB cases and half a million additional deaths in the period 2025–35 in the absence of USAID funding; with concurrent reductions in the Global Fund, these numbers could rise to 5 million additional cases and 2 million deaths.⁴⁷

In addition to HIV programmes, emerging evidence has revealed that cuts to global funding are directly affecting TB control services. A rapid survey across 52 countries, primarily in Africa, found that 65% of national TB programmes (NTPs) reported staff layoffs, reduced capacity for case detection, disruptions to laboratory operations and expansion, as well as decreased

	Risk-deleted DALYs	Risk-deleted DALY rate per 100 000	Risk-deleted YLLs	Risk-deleted YLL rate per 100 000	Risk-deleted deaths	Risk-deleted death rate per 100 000
Global						
TB	34 900 000 (27 800 000 to 43 800 000)	433.2 (344.6 to 542.5)	30 000 000 (23 300 000 to 38 300 000)	372.4 (289.3 to 475.0)	768 000 (592 000 to 970 000)	9.5 (7.3 to 12.0)
Drug-susceptible TB	31 800 000 (24 500 000 to 40 400 000)	394.4 (304.2 to 500.6)	27 100 000 (19 600 000 to 35 300 000)	336.4 (243.1 to 437.4)	691 000 (509 000 to 888 000)	8.6 (6.3 to 11.0)
MDR-TB	3 120 000 (1 030 000 to 7 290 000)	38.7 (12.7 to 90.4)	2 900 000 (921 000 to 6 670 000)	36.0 (11.4 to 82.7)	77 200 (23 400 to 183 000)	1.0 (0.3 to 2.3)
Central Europe, eastern Europe, and central Asia						
TB	309 000 (250 000 to 366 000)	73.0 (59.2 to 86.6)	273 000 (223 000 to 329 000)	64.5 (52.6 to 77.7)	7030 (5790 to 8400)	1.7 (1.4 to 2.0)
Drug-susceptible TB	195 000 (113 000 to 268 000)	46.1 (26.7 to 63.3)	168 000 (92 000 to 235 000)	39.7 (21.7 to 55.6)	4350 (2540 to 6100)	1.0 (0.6 to 1.4)
MDR-TB	114 000 (57 000 to 184 000)	26.9 (13.5 to 43.5)	105 000 (51 400 to 171 000)	24.8 (12.2 to 40.5)	2680 (1340 to 4250)	0.6 (0.3 to 1.0)
High income						
TB	136 000 (102 000 to 162 000)	12.3 (9.2 to 14.7)	117 000 (88 100 to 141 000)	10.6 (8.0 to 12.8)	7310 (5650 to 8710)	0.7 (0.5 to 0.8)
Drug-susceptible TB	132 000 (99 200 to 158 000)	11.9 (9.0 to 14.3)	113 000 (86 300 to 136 000)	10.2 (7.8 to 12.4)	7070 (5490 to 8520)	0.6 (0.5 to 0.8)
MDR-TB	4290 (1700 to 9450)	0.4 (0.2 to 0.9)	3970 (1440 to 8990)	0.4 (0.1 to 0.8)	238 (93 to 552)	0.0 (0.0 to 0.0)
Latin America and Caribbean						
TB	864 000 (681 000 to 1 070 000)	146.0 (114.9 to 181.3)	757 000 (575 000 to 963 000)	127.9 (97.1 to 162.7)	19 100 (15 200 to 23 400)	3.2 (2.6 to 3.9)
Drug-susceptible TB	813 000 (608 000 to 1 010 000)	137.3 (102.7 to 170.2)	710 000 (530 000 to 896 000)	119.9 (89.5 to 151.3)	17 800 (13 600 to 22 100)	3.0 (2.3 to 3.7)
MDR-TB	51 100 (21 100 to 116 000)	8.6 (3.6 to 19.6)	47 600 (19 000 to 110 000)	8.0 (3.2 to 18.6)	1290 (530 to 2990)	0.2 (0.1 to 0.5)
North Africa and Middle East						
TB	605 000 (403 000 to 907 000)	94.8 (63.2 to 142.1)	484 000 (290 000 to 782 000)	75.8 (45.5 to 122.6)	12 000 (7300 to 20 500)	1.9 (1.1 to 3.2)
Drug-susceptible TB	569 000 (381 000 to 857 000)	89.2 (59.8 to 134.3)	451 000 (275 000 to 740 000)	70.7 (43.2 to 115.9)	11 300 (6990 to 19 300)	1.8 (1.1 to 3.0)
MDR-TB	36 000 (11 900 to 89 700)	5.6 (1.9 to 14.1)	32 300 (9560 to 84 500)	5.1 (1.5 to 13.2)	769 (237 to 1900)	0.1 (0.0 to 0.3)
South Asia						
TB	12 500 000 (10 100 000 to 15 800 000)	657.8 (533.7 to 834.5)	10 700 000 (8 320 000 to 14 100 000)	564.9 (439.5 to 742.2)	314 000 (245 000 to 413 000)	16.6 (13.0 to 21.8)
Drug-susceptible TB	10 800 000 (6 930 000 to 14 100 000)	569.7 (366.2 to 746.9)	9 150 000 (5 640 000 to 12 500 000)	483.4 (298.1 to 661.8)	268 000 (164 000 to 371 000)	14.2 (8.7 to 19.6)
MDR-TB	1 670 000 (240 000 to 4 320 000)	88.1 (12.7 to 228.2)	1 540 000 (214 000 to 4 020 000)	81.5 (11.3 to 212.3)	45 500 (6320 to 122 000)	2.4 (0.3 to 6.4)
Southeast Asia, east Asia, and Oceania						
TB	5 270 000 (4 350 000 to 6 360 000)	238.3 (196.9 to 287.9)	4 210 000 (3 270 000 to 5 280 000)	190.4 (148.1 to 239.2)	121 000 (95 000 to 149 000)	5.5 (4.3 to 6.8)
Drug-susceptible TB	5 030 000 (4 070 000 to 6 080 000)	227.9 (184.4 to 275.1)	4 000 000 (3 070 000 to 5 010 000)	181.2 (138.9 to 226.9)	115 000 (88 500 to 141 000)	5.2 (4.0 to 6.4)
MDR-TB	230 000 (93 400 to 477 000)	10.4 (4.2 to 21.6)	204 000 (76 900 to 440 000)	9.2 (3.5 to 19.9)	6280 (2320 to 13 400)	0.3 (0.1 to 0.6)
Sub-Saharan Africa						
TB	15 300 000 (10 500 000 to 21 700 000)	1268.4 (868.3 to 1796.0)	13 500 000 (8 600 000 to 19 600 000)	1119.0 (712.7 to 1620.7)	288 000 (187 000 to 410 000)	23.8 (15.5 to 34.0)
Drug-susceptible TB	14 300 000 (9 570 000 to 20 100 000)	1183.7 (793.2 to 1666.1)	12 500 000 (7 900 000 to 18 200 000)	1039.0 (654.4 to 1508.9)	267 000 (173 000 to 387 000)	22.1 (14.3 to 32.1)
MDR-TB	1 020 000 (330 000 to 2 470 000)	84.6 (27.4 to 204.6)	965 000 (296 000 to 2 360 000)	80.0 (24.5 to 195.2)	20 500 (6440 to 50 100)	1.7 (0.5 to 4.2)

To be consistent with GBD classification, drug-susceptible TB includes all tuberculosis cases not meeting the criteria for MDR-TB. While primarily defined as being susceptible to isoniazid and rifampicin, this reporting category also encompasses other non-MDR drug-resistant strains, such as isoniazid-monoresistant TB. The risk-deleted estimates reflect the combined removal of smoking, alcohol use, and high fasting plasma glucose combined. MDR-TB estimates include XDR-TB. DALY=disability-adjusted life-year. GBD=Global Burden of Diseases, Injuries, and Risk Factors Study. MDR=multidrug-resistant. TB=tuberculosis. YLL=years of life lost. XDR=extensively drug resistant.

Table 2: Risk-deleted DALYs, YLLs, and deaths with corresponding all-age rates by drug resistance status in 2023 by GBD super-region

	2015 incidence rate per 100 000	2023 incidence rate per 100 000	Percentage change in incidence rate	Absolute difference in incidence rate	2015 deaths	2023 deaths	Percentage change in mortality	Absolute difference in mortality
Global								
All-form TB	139.8 (122.4 to 157.2)	112.9 (99.6 to 127.5)	-19.2% (-20.5 to -17.8)*	-26.9 (-31.4 to -22.6)	1 580 000 (1 340 000 to 1 940 000)	1 220 000 (975 000 to 1 490 000)	-22.6% (-35.7 to -4.7)*	-356 000 (-663 000 to -71 700)
Drug-susceptible TB	118.7 (104.0 to 133.1)	97.5 (84.5 to 110.0)	-17.9% (-21.2 to -15.5)*	-21.2 (-26.2 to -17.0)	1 160 000 (924 000 to 1 490 000)	908 000 (671 000 to 1 140 000)	-21.7% (-36.7 to 0.1)	-252 000 (-497 000 to 871)
MDR-TB	6.5 (4.2 to 12.7)	5.8 (2.5 to 13.4)	-11.4% (-48.7 to 41.2)	-0.7 (-3.1 to 2.7)	118 000 (52 900 to 252 000)	102 000 (31 700 to 238 000)	-13.6% (-49.7 to 33.0)	-16 100 (-73 200 to 39 100)
HIV/AIDS-related drug-susceptible TB	14.1 (12.4 to 15.9)	9.3 (8.2 to 10.6)	-33.9% (-36.8 to -32.1)*	-4.8 (-5.5 to -4.2)	273 000 (196 000 to 348 000)	190 000 (125 000 to 253 000)	-30.1% (-40.1 to -19.8)*	-82 100 (-112 000 to -46 600)
HIV/AIDS-related MDR-TB	0.5 (0.3 to 0.7)	0.3 (0.2 to 0.7)	-27.2% (-55.1 to 19.9)	-0.1 (-0.3 to 0.1)	25 100 (11 100 to 48 000)	19 600 (6 840 to 40 500)	-21.9% (-51.2 to 24.1)	-5490 (-19 100 to 7710)
Central Europe, eastern Europe, and central Asia								
All-form TB	72.1 (60.0 to 86.1)	50.1 (41.8 to 60.4)	-30.5% (-36.6 to -22.7)*	-22.0 (-28.8 to -15.7)	26 900 (25 300 to 28 600)	14 800 (13 900 to 16 000)	-44.9% (-48.8 to -41.2)*	-12 100 (-13 600 to -10 600)
Drug-susceptible TB	46.8 (35.8 to 57.4)	33.4 (24.8 to 42.1)	-28.7% (-43.5 to -15.7)*	-13.4 (-20.3 to -5.8)	13 900 (9 270 to 17 900)	8230 (5 410 to 10 800)	-40.7% (-58.7 to -19.7)*	-5640 (-8830 to -1780)
MDR-TB	22.3 (13.7 to 38.1)	14.8 (7.7 to 27.2)	-33.7% (-65.7 to 16.5)	-7.5 (-20.6 to 3.1)	10 000 (5950 to 14 400)	5100 (2690 to 7940)	-49.1% (-68.7 to -22.0)*	-4920 (-8630 to -1790)
HIV/AIDS-related drug-susceptible TB	2.2 (1.4 to 2.9)	1.4 (0.8 to 2.0)	-34.9% (-57.9 to -9.5)*	-0.8 (-1.3 to -0.2)	1400 (609 to 2480)	729 (305 to 1260)	-47.9% (-71.4 to -14.5)*	-671 (-1450 to -112)
HIV/AIDS-related MDR-TB	0.8 (0.5 to 1.5)	0.5 (0.2 to 1.0)	-39.5% (-69.8 to 9.8)	-0.3 (-0.8 to 0.1)	1590 (727 to 2850)	760 (294 to 1400)	-52.2% (-70.5 to -27.9)*	-829 (-1660 to -290)
High income								
All-form TB	10.3 (9.0 to 11.8)	8.2 (7.1 to 9.4)	-20.7% (-22.7 to -18.5)*	-2.1 (-2.5 to -1.8)	13 100 (11 400 to 14 700)	13 100 (11 200 to 14 600)	0.0% (-6.7 to 6.8)	5 (-911 to 843)
Drug-susceptible TB	9.5 (8.3 to 10.9)	7.5 (6.5 to 8.6)	-21.2% (-23.2 to -18.9)*	-2.0 (-2.3 to -1.7)	11 000 (9330 to 12 400)	11 300 (9380 to 12 800)	2.1% (-5.0 to 9.4)	231 (-541 to 1050)
MDR-TB	0.2 (0.1 to 0.3)	0.1 (0.1 to 0.3)	-18.8% (-46.8 to 15.1)	0.0 (-0.1 to 0.0)	406 (184 to 855)	390 (166 to 856)	-4.0% (-37.2 to 37.3)	-16 (-208 to 156)
HIV/AIDS-related drug-susceptible TB	0.6 (0.6 to 0.7)	0.5 (0.5 to 0.6)	-14.0% (-16.4 to -11.7)*	-0.1 (-0.1 to -0.1)	1610 (1060 to 2150)	1420 (936 to 1850)	-12.3% (-19.3 to -4.4)*	-199 (-381 to -54)
HIV/AIDS-related MDR-TB	0.0 (0.0 to 0.0)	0.0 (0.0 to 0.0)	-15.4% (-55.2 to 52.6)	0.0 (0.0 to 0.0)	74 (27 to 163)	64 (20 to 152)	-13.5% (-53.0 to 61.3)	-10 (-52 to 46)
Latin America and Caribbean								
All-form TB	41.1 (35.4 to 47.3)	41.3 (35.2 to 48.1)	0.6% (-2.2 to 3.4)	0.3 (-0.8 to 1.6)	33 100 (27 200 to 38 500)	36 000 (30 900 to 41 200)	8.9% (-0.9 to 21.0)	2930 (-346 to 6610)
Drug-susceptible TB	34.2 (29.5 to 39.3)	34.7 (29.7 to 40.3)	1.5% (-2.7 to 5.0)	0.5 (-0.9 to 1.7)	22 400 (18 800 to 26 200)	25 400 (21 200 to 29 600)	13.6% (0.6 to 28.7)*	3060 (115 to 6400)
MDR-TB	1.2 (0.8 to 2.1)	1.4 (0.7 to 3.5)	20.8% (-34.7 to 138.4)	0.2 (-0.4 to 2.0)	1330 (576 to 2710)	1860 (750 to 4480)	40.0% (-21.7 to 178.2)	531 (-402 to 2190)
HIV/AIDS-related drug-susceptible TB	5.6 (4.8 to 6.4)	5.1 (4.3 to 5.9)	-9.1% (-14.2 to -5.7)*	-0.5 (-0.7 to -0.3)	8860 (5820 to 12 200)	8130 (5390 to 11 200)	-8.3% (-18.7 to 3.7)	-732 (-1650 to 222)
HIV/AIDS-related MDR-TB	0.1 (0.1 to 0.3)	0.1 (0.0 to 0.4)	10.2% (-51.0 to 132.4)	0.0 (-0.1 to 0.2)	484 (138 to 1210)	558 (146 to 1550)	15.5% (-44.8 to 163.5)	75 (-353 to 693)
North Africa and Middle East								
All-form TB	41.9 (35.4 to 48.4)	35.4 (30.2 to 40.8)	-15.4% (-17.7 to -12.9)*	-6.4 (-8.0 to -5.1)	19 200 (12 800 to 30 600)	16 800 (11 200 to 26 600)	-12.3% (-35.1 to 22.4)	-2360 (-8550 to 4280)

(Table 3 continues on next page)

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	2015 incidence rate per 100 000	2023 incidence rate per 100 000	Percentage change in incidence rate	Absolute difference in incidence rate	2015 deaths	2023 deaths
					Percentage change in mortality	Absolute difference in mortality
South Asia						
Drug-susceptible TB	39.5 (33.5 to 45.7)	33.5 (28.5 to 38.7)	-15.1% (-18.3 to -12.0)*	-6.0 (-7.5 to -4.4)	16 800 (10 900 to 27 100)	14 800 (9840 to 23 700)
MDR-TB	1.1 (0.7 to 1.7)	1.0 (0.5 to 2.2)	-15.9% (-63.4 to 67.1)	-0.2 (-1.0 to 0.9)	1170 (468 to 2410)	1000 (311 to 2490)
HIV/AIDS-related drug-susceptible TB	1.2 (1.1 to 1.4)	0.9 (0.8 to 1.1)	-23.6% (-26.2 to -20.0)*	-0.3 (-0.3 to -0.2)	1100 (620 to 1820)	890 (474 to 1560)
HIV/AIDS-related MDR-TB	0.0 (0.0 to 0.0)	0.0 (0.0 to 0.0)	-21.2% (-70.9 to 84.5)	0.0 (0.0 to 0.0)	64 (19 to 153)	55 (15 to 155)
Southeast Asia, east Asia, and Oceania						
All-form TB	213.0 (184.8 to 246.1)	158.3 (138.0 to 180.2)	-25.7% (-28.0 to -23.1)*	-54.7 (-66.9 to -44.2)	590 000 (505 000 to 709 000)	426 000 (336 000 to 550 000)
Drug-susceptible TB	193.8 (162.4 to 224.3)	143.4 (118.3 to 166.7)	-26.0% (-31.4 to -22.4)*	-50.4 (-62.3 to -39.5)	494 000 (370 000 to 629 000)	353 000 (222 000 to 484 000)
MDR-TB	12.9 (3.4 to 36.0)	11.9 (1.9 to 31.6)	-7.6% (-50.6 to 45.4)	-1.0 (-6.5 to 4.8)	67 800 (15 200 to 184 000)	59 700 (8870 to 162 000)
HIV/AIDS-related drug-susceptible TB	5.9 (4.9 to 6.9)	2.8 (2.1 to 3.2)	-53.2% (-57.4 to -50.6)*	-3.1 (-3.7 to -2.6)	23 900 (15 500 to 32 600)	11 400 (6440 to 16 400)
HIV/AIDS-related MDR-TB	0.3 (0.1 to 0.9)	0.2 (0.0 to 0.5)	-45.8% (-70.3 to -14.2)*	-0.1 (-0.5 to 0.0)	3820 (729 to 10 300)	2230 (309 to 6550)
Southeast Asia, east Asia, and Oceania						
All-form TB	110.1 (98.6 to 121.0)	96.1 (86.4 to 105.5)	-12.7% (-14.0 to -11.1)*	-14.0 (-16.4 to -11.5)	251 000 (212 000 to 306 000)	195 000 (155 000 to 235 000)
Drug-susceptible TB	102.5 (91.2 to 112.5)	89.9 (79.9 to 99.4)	-12.2% (-15.7 to -9.8)*	-12.5 (-16.0 to -9.6)	220 000 (182 000 to 273 000)	172 000 (138 000 to 209 000)
MDR-TB	3.1 (2.0 to 5.9)	2.9 (1.2 to 7.6)	-4.1% (-8.0 to 91.7)	-0.1 (-2.3 to 3.1)	12 000 (6060 to 22 900)	9810 (3630 to 21 200)
HIV/AIDS-related drug-susceptible TB	4.5 (4.0 to 4.9)	3.2 (2.8 to 3.5)	-29.5% (-32.5 to -27.1)*	-1.3 (-1.5 to -1.1)	17 300 (12 300 to 22 400)	11 600 (7690 to 15 600)
HIV/AIDS-related MDR-TB	0.1 (0.1 to 0.2)	0.1 (0.0 to 0.2)	-29.6% (-61.3 to 15.0)	0.0 (-0.1 to 0.0)	1670 (720 to 3280)	1080 (397 to 2090)
Sub-Saharan Africa						
All-form TB	359.2 (312.0 to 407.1)	266.5 (230.0 to 302.2)	-25.8% (-27.3 to -24.1)*	-92.7 (-104.4 to -78.7)	643 000 (479 000 to 851 000)	518 000 (362 000 to 670 000)
Drug-susceptible TB	266.3 (226.2 to 304.5)	208.6 (178.7 to 239.4)	-21.7% (-24.8 to -18.7)*	-57.7 (-68.7 to -46.5)	381 000 (261 000 to 556 000)	323 000 (212 000 to 461 000)
MDR-TB	9.1 (6.1 to 16.4)	8.0 (3.8 to 18.1)	-11.9% (-58.0 to 59.1)	-1.1 (-8.2 to 5.3)	25 500 (9820 to 62 000)	24 400 (7760 to 61 300)
HIV/AIDS-related drug-susceptible TB	81.4 (70.9 to 91.7)	48.3 (42.2 to 54.4)	-40.7% (-43.3 to -38.9)	-33.2 (-38.0 to -28.9)	218 000 (153 000 to 280 000)	156 000 (101 000 to 208 000)
HIV/AIDS-related MDR-TB	2.4 (1.6 to 3.7)	1.6 (0.9 to 3.3)	-30.2% (-62.4 to 18.6)	-0.7 (-2.2 to 0.5)	17 400 (7410 to 36 400)	14 800 (5060 to 31 800)

Data in parentheses are 95% uncertainty intervals. All-form TB represents the sum of four subsequent rows (drug-susceptible TB, MDR-TB, HIV/AIDS-related drug-susceptible TB, and HIV/AIDS-related MDR-TB). For clarification, drug-susceptible TB and MDR-TB refer to TB among people without HIV. To be consistent with GBD classification, drug-susceptible TB includes all TB cases not meeting the criteria for MDR-TB. While primarily defined as being susceptible to isoniazid and rifampicin, this reporting category also encompasses other non-MDR drug-resistant strains such as isoniazid-monoresistant TB. MDR-TB estimates include XDR-TB. GBD=Global Burden of Diseases, Injuries, and Risk Factors Study. MDR=multidrug-resistant TB=tuberculosis. XDR=extensively drug resistant. *Interval does not cross the zero threshold, denoting statistical significance.

Table 3: Percentage changes in all-form tuberculosis incidence rates and number of deaths from 2015 to 2023 by GBD super-region, stratified by drug-resistance and HIV status

financial resources for patient outreach and adherence monitoring.¹⁴ Similar disruptions were also observed in several countries in southeast Asia, where NTPs have historically relied heavily on foreign aid.⁴⁸ Such disruptions, even if brief, are predicted to cause substantial increases in TB incidence and excess mortality in high-burden countries.^{49,50} Mechanistically, these disruptions could drive a resurgence of TB, especially MDR-TB, mirroring the post-Soviet collapse during the 1990s, when drug supply interruptions and incomplete treatment fuelled drug resistance.^{51,52} Future studies should further incorporate long-term projections of TB burden by drug-resistance status to more fully assess the potential consequences of reductions in global funding. Overall, current modelling studies outline a range of plausible scenarios, ranging from rapid recovery if funding is reinstated, to stalled progress, to a worst-case scenario in which a prolonged funding contraction would erase much of the gains of the past two decades.^{47–50}

MDR-TB continues to pose one of the most arduous challenges in global TB prevention efforts because of its limited treatment response and higher fatality compared with drug-susceptible TB. The 2025 WHO Global TB report identified resistance to rifampicin, the most effective first-line TB drug, as a major concern for TB prevention.⁵³ In 2023, we estimated there were 466 000 (95% UI 198 000–1 080 000) incident cases of MDR-TB and 102 000 (31 700–238 000) deaths. MDR-TB is extremely difficult to treat, mandating use of more toxic agents with lower treatment success rates,⁵⁴ and hence imposing a disproportionate global mortality burden. The 3·96 million (1·31–9·01) DALYs caused by MDR-TB were disproportionately driven by premature mortality (3·68 million [1·18–8·31] YLLs) as opposed to disability (277 000 [106 000–662 000] YLDs). This pronounced imbalance highlights that MDR-TB remains an overwhelmingly fatal disease. Drug-resistance patterns coupled with delayed diagnosis, treatment intolerance, and non-adherence, as well as under-resourced NTPs and health systems, further contribute to the high lethality of MDR-TB.⁵⁵ HIV is also an important cofactor in MDR-TB progress and prognosis because HIV co-infection amplifies diagnostic and treatment challenges.⁵⁶ Compared to individuals without HIV, people living with HIV experience faster progression from infection to active TB disease and poorer treatment success when drug resistance is present.^{56,57} Compounding this challenge are overlapping drug–drug interactions, particularly involving bedaquiline, linezolid, and ART, which lead to treatment discontinuation, thereby perpetuating the cycle of increased drug resistance and community spread.⁵⁸ This interaction might explain the continued high MDR-TB mortality burden in sub-Saharan Africa despite substantial ART scale-up and advances in HIV care. Broadening rapid diagnosis and ensuring treatment completion are crucial for reducing the MDR-TB burden.⁵⁵

New treatment frameworks for MDR-TB offer promise to transform a disease once considered nearly incurable into one that is increasingly manageable and curable. The 2024 American Thoracic Society/US Centers for Disease Control and Prevention/European Respiratory Society/Infectious Diseases Society of America guidelines mark a major inflection point in the treatment of both drug-susceptible and MDR-TB.⁵⁹ For drug-susceptible TB, the guidelines endorse shorter, all-oral regimens, including a 4-month course of isoniazid, rifapentine, pyrazinamide, and moxifloxacin, replacing the traditional 6-month regimen. For MDR-TB and selected pre-XDR TB cases, the guidelines recommend 6-month bedaquiline-pretomanid-linezolid-moxifloxacin and bedaquiline-pretomanid-linezolid regimens based on clinical trial evidence demonstrating treatment success exceeding 85%—a substantial improvement over historical 18–24-month regimens that relied on toxic injectable agents. These advances have the potential to shorten infectious periods, reduce treatment non-adherence, and improve overall survival. However, the population-level impact of these therapeutic breakthroughs hinges on robust implementation strategies. Increasing rates of bedaquiline resistance have already been documented in settings where diagnostic capacity is limited and treatment adherence is suboptimal.^{60–62} Left unchecked, the emergence of new resistant strains threatens to subvert the efficacy of the very regimens designed to halt its spread.

Strengthening laboratory capacity for rapid drug-resistance testing and ensuring equitable access to novel TB drugs must underpin scale-up of clinical innovations. WHO recommends rapid molecular diagnostics as the initial test for all individuals with suspected TB,⁶³ with rifampicin-resistance testing for confirmed cases and fluoroquinolone-resistance testing among those with rifampicin-resistant disease. Despite these recommendations, approximately 60% of MDR-TB cases remained undetected globally in 2022.⁶⁴ Conversely, the high costs of essential medicines—namely, pretomanid and bedaquiline—create inequity in access, with patent-related and licensing constraints further limiting availability in resource-constrained settings.⁵⁷ Recent price reductions announced by the Stop TB Partnership's Global Drug Facility—25% for pretomanid and 68% for bedaquiline—have reduced the cost of a full bedaquiline-pretomanid-linezolid-moxifloxacin regimen to less than US\$300 for the first time.^{57,58} However, these reductions might not translate into higher treatment coverage in the context of ongoing funding constraints and supply-chain vulnerabilities.

Sustained investment in research and innovations is essential to achieving the WHO End TB agenda, particularly for the development of new vaccines, diagnostics, and therapeutics. As of August, 2025, WHO reported 18 TB vaccine candidates and 29 TB drugs in clinical development. However, these efforts are

increasingly threatened by reductions in research funding.⁶⁵ More than a third of TB research funding comes from the US National Institutes of Health (NIH),⁶⁶ and the 2025 US budgetary contractions resulted in the termination of more than 14% of infectious disease clinical trials between February and August of that year.⁶⁷ While the US Consolidated Appropriations Act restored NIH funding with a modest increase,⁴⁵ the reinstatement of previously terminated TB research awards remains uncertain. Nonetheless, the Act establishes essential legislative guardrails that protect the biomedical research enterprise from erratic disruptions, offering temporary stability.⁶⁸

Although advances in prevention and therapeutics are essential, mitigating modifiable risk factors remains an underused but impactful strategy for global TB prevention. Our results indicate that eliminating smoking, alcohol use, and high fasting plasma glucose could reduce mortality and YLLs for both all-form TB and MDR-TB by more than 30%. The impacts of these risks are reflected in large regional disparities. In Lesotho, for example, where smoking prevalence is more than 2·5 times the average for sub-Saharan Africa (GBD; Global Health Data Exchange),⁶⁹ the HIV-related TB mortality rate exceeds the regional mean by more than eight-fold. In Latin America and the Caribbean, stalled TB progress, driven in part by socioeconomic instability⁷⁰ and rising incarceration,⁷¹ is further compounded by rapidly increasing diabetes prevalence,⁷² which has more than doubled in parts of the region since 1990.⁷³ Addressing these syndemics will require coordinated, multisectoral action to limit the impacts of unhealthy commodities, including tobacco, alcohol, and ultra-processed, high-sugar foods. Proven, cost-effective interventions exist^{74,75} and could represent cornerstones of sustainable TB prevention.

Compared with WHO 2025 estimates, global TB mortality estimates for GBD 2023 were similar at the global level (1·22 million [95% UI 0·98–1·49] for GBD 2023 vs 1·27 million [1·17–1·38] for WHO 2025). By contrast, larger differences were observed for TB incidence (9·11 million [8·04–10·3] for GBD 2023 vs 10·8 million [9·9–11·5] for WHO 2025). These discrepancies are likely to arise from methodological differences, particularly in data-sparse settings. WHO estimates rely primarily on prevalence surveys or case notifications adjusted for underdiagnosis, whereas GBD uses a statistical triangulation approach integrating mortality, prevalence, notifications, and incidence based on mortality-to-incidence ratios through the DisMod-MR 2.1 tool, which ensures internal consistency across these parameters. As a result, GBD estimates are generally less directly anchored to notification data than WHO estimates and might also show different temporal trends. Further discrepancies stem from WHO's use of modelling assumptions to adjust for service disruptions due to the COVID-19 pandemic. In GBD 2023, we did

not modify the underlying modelling framework for the 2020–21 period to avoid introducing assumptions; instead, we incorporated newly available cause-of-death and prevalence survey data, including those for the pandemic years, within the standard estimation pipeline. Recent data have suggested more limited impacts of the COVID-19 pandemic on TB than initially projected by some models. Continued data collection and analysis are necessary to fully understand the long-term impacts of the COVID-19 pandemic on TB.

Our findings should be interpreted in light of the following limitations. First, data availability varied across countries, demographics, and years. Where direct data were absent, statistical models were applied, leveraging covariate information and spatial and temporal relationships to predict missing values. Despite our best efforts to select the most robust models, high levels of uncertainty remain. Data scarcity issues are also pronounced in MDR-TB burden estimation. The wide uncertainty intervals for MDR-TB estimates reflect the limited availability of time-series data, especially for many countries in south Asia, southeast Asia, east Asia, central Asia, sub-Saharan Africa, and north Africa and the Middle East; these substantial intervals are particularly evident in percentage change estimates, as relative measures can magnify underlying model uncertainty when baseline rates are low. Similarly, owing to sparse and highly variable country-level data on XDR-TB, proportions were estimated at the GBD super-region level to improve model stability. However, this approach could obscure heterogeneity within super-regions and variation across countries. Additionally, we applied a single global relative risk of death for MDR-TB compared with drug-susceptible TB, which might not capture variation across settings with differing health-system quality and capacity. Strengthening drug-resistance testing capacity is therefore essential, not only to optimise treatment strategies but also to deepen our understanding of local, regional, and global resistance patterns and disease evolution. Enhanced data availability would also enable more granular estimation of TB burden in GBD, including disaggregation of drug-susceptible TB to account for monoresistant forms. Second, verbal autopsy data were used in countries without vital registration data. However, verbal autopsy has suboptimal sensitivity for identifying TB deaths and for differentiating HIV deaths from deaths resulting from HIV and TB co-infection. In settings with high HIV-prevalence, verbal autopsy data were excluded and estimates were derived from modelling instead. Third, data sparseness also posed a challenge in our efforts to triangulate different metrics through our disease modelling step. Specifically, the relationships between prevalence and mortality are largely driven by locations with robust data. Expanding surveillance to capture both prevalence and mortality in data-sparse settings would bolster our ability to monitor the epidemic's trajectory.

For more on the Global Health Data Exchange see <https://ghdx.healthdata.org/series/global-burden-disease-ghd>

Finally, our analysis was limited to three major TB risk factors. While the GBD framework considers 88 modifiable risk factors, not all established drivers are currently included; for example, undernutrition or underweight, air pollution, HIV infection, and social determinants of TB have yet to be incorporated. As a result, our risk-deleted estimates are likely to underestimate the total modifiable TB burden. Furthermore, we did not quantify the interaction between HIV and these risk factors on TB burden.

Tuberculosis remains a formidable global health threat. Although substantial progress was achieved, particularly in the early 2000s, recent stagnation and resurgence underscore how fragile these gains remain. The COVID-19 pandemic showed that controlling infectious diseases is not a unilateral endeavour, but a global commitment requiring sustained collective investment. Amid growing uncertainty surrounding US global health commitments, renewed international solidarity is essential to protect past advances and accelerate efforts to prevent millions of avoidable TB deaths.

GBD 2023 TB HIV Collaborators

Jorge R Ledesma*, Meixin Zhang*, Mei Fong Liew*, Regina-Mae Villanueva Dominguez, Sophie Mei Lin Whitehart, Megan Verma, Jianing Ma, Amanda Movo, Huong Thi Chu, Samuel M Ostroff, Jiawei He, Abdullah , Hatem A Eltaly, Ukachukwu O Abaraogu, Belayneh Jejaw Abate, Mazhar Abbas, Samar Abd ElHafeez, Emad M Abdallah, Sherief Abd-Elsalam, Siddig Ibrahim Abdelwahab, Wakgari Mosisa Abdisa, Ali Abdolizadeh, Hassan A. Abdou, Meriem Abdoun, Arman Abdous, Jeza Muhammad Abdul Aziz, Deldar Morad Abdullah, Yasir M Abdulateef, Rashad Abdul-Ghani, Auwal Abdullahi, Abdulrakib Abdulrahim, Rezheen Fatah Abdulrahman, Zirak Ahmed Abdulrahman, Sadik Abdulwehab, Habtamu Abebe Getahun, Asrat Agalu Abejew, Syed Hani Abidi, Olifan Zewdie Abil, Aarailym Abilbayeva, Olumide Abiodun, Richard Gyan Aboagye, Shima A Aboelnaga, Hassan Abolhassani, Abdullahi Tunde Aborode, Nagah M Abourashed, Amr Selim Abu Lila, Bilyaminu Abubakar, Nermeen Abu-Elala, Rana Kamal Abu-Farha, Eman Abu-Gharbieh, Ali Abuhaliema, Sawsan H Abuhammad, Ahmad Y Abuhelwa, Hana J Abukhadajah, Aseel Aburub, Mai Abdel Haleem Abusalah, Anirudh Balakrishna Acharya, Krishna Prasad Acharya, Swetha Acharya, Abdu A Adamu, Lawan Hassan Adamu, Nuhu Lawan Adamu, Wondimnew Desalegn Addis, Isaac Yeboah Addo, Tajudeen Adesanmi Adebisi, Babatope Oluwadamilare Adebiji, Oluwawemimo Oluseun Adebowale, Kamoru Ademola Adedokun, Oluwatobi E Adegbile, Victor Adekanmbi, Saheed Ayodeji Adekola, Olumide Thomas Adeleke, Makinde Adebayo Adeniyi, Abiola Victor Adepoju, Ridwan Olamilekan Adesola, Olatunji O Adetokunboh, Charles Oluwaseun Adetunji, Juliana Bunmi Adetunji, Ripon Kumar Adhikary, Usha Adiga, Molalign Aligaz Adisu, Belete Muluadam Admassie, Mohd Adnan, Prince Owusu Adoma, Obed Adonteng-Kissi, Aanuoluwapo Adeyimika Afolabi, Ebenezer Afrifa-Yamoah, Muhammad Sohail Afzal, Saira Afzal, Gizachew Beykaso Agafari, Dhiraj Motilal Agarwal, Navidha Aggarwal, Vijay K Aggarwal, Mahdi Aghaali khani, Feleke Doyore Agide, Percival Delali Agordoh, Williams Agyemang-Duah, Abdul Momin Rizwan Ahmad, Adamu Adamu Ahmad, Adnan Ahmad, Hassan Yusuf Ahmad, Irfan Ahmad, Khurshid Ahmad, Munir Ahmad, Rabbiya Ahmad, Saheem Ahmad, Sajjad Ahmad, Tauseef Ahmad, Ali Ahmed, Ayman Ahmed, Jivan Qasim Ahmed, Khalid A Ahmed, Mehrunnisha Sharif Ahmed, Meqdad Saleh Ahmed, Mohamed Sherif Ali Ahmed, Mohammed Ahmed, Mushood Ahmed, Naveed Ahmed, Shahzaib Ahmed, Syed Anees Ahmed,

Akmaral Ainakulova, Gizachew Tadesse Akalu, Ruslan Akhmedullin, Neda Akhouni, Akil Akhtar, Muhammad Nadeem Akhtar, Muzamil Akhtar, Oladipo Vincent Akinmade, Karolina Akinosoglou, Damilare Michael Akintunde, Hammad Akram, Amer Hashim Al Ani, Salah Al Awaidey, Syed Mahfuz Al Hasan, Khalid Awad Al Kubaisi, Duaa Fayiz Al Maghaireh, Nasser Al salmi, Omar Ali Mohammed Al Zaabi, Mohammad Ahmmad Mahmoud Al Zoubi, Tariq Al Zoubi, Khurshid Alam, Mohammad Khurshed Alam, Nazmul Alam, Fahad Mashhour Alanezi, Amani N. Alansari, Fahmi Y Al-Ashwal, Seyed Mohammad Amin Alavi, Anas Albashaish, Mohammed Albashtawy, Mohammad T AlBataineh, M.S. Sultan Al-Buriah, Abdulleh Mastour Aldhahir, Embay Amare Alemseged, Kefyalew Addis Alene, Ali M Alfalki, Fahad D Algahtani, Abdelazeem M Alghammal, Mohammed Ridha Algethami, Saeed M. M Alghamdi, Inayat Ali, Irfan Ali, Kamran Ali, Liaqat Ali, Mohammad Daud Ali, Mohammad Shahid Ali, Mohammed Usman Usman Ali, Rafat Ali, Syed Shujait Ali, Rawaa Ali Hussein Saleh, Montaha Al-Iede, Sheikh Mohammad Alif, Hamid Alinejad Rokny, Mohammed Alissa, Alaa A A Aljabali, Samah W Al-Jabi, Ahmed Aljabri, Moath Saleh Aljohani, Syed Mohamed Aljunid, Hazim Alkousheh, Sameer Abdulmalik Alkubati, Kasim Allel, Khaled S. Allemailem, Adrien Allorant, Mohammed Z. Allouh, Sabah Al-Marwani, Ahmad Abdulaziz Almatroudi, Joseph Uy Almazan, Hesham M Al-Mekhlafi, Khalid Almuqheed, Mohammad Minwer Alnaeem, Khaldoun Aied Alnawafleh, Hasan Yaser Alniss, Mushtak Talib Salih Al-Ouqaili, Mohammad A Y Alqudah, Mohammad R Alqudimat, Ahmad Rajeh Al-Qudimat, Osamah Matar Alrasheedy, Mousa Alreshidi, Bandar Sulaiman Alsaif, Najim Z Alshahrani, Ali Salman Al-Shami, Bushra Alshammari, Mamdouh Alshammari, Hamad Alshetaiwi, Yazed Saleh Alsowaida, Zaid Altaany, Awais Altaf, Jaffar A Al-Tawfiq, Kassaye D. Demeke Altaye, Malik A Althobiani, Anwar Noor Althubiani, Dalia Altwalbeh, Nelson Alvis-Guzman, Nelson J Alvis-Zakzuk, Yaser Mohammed Al-Worafi, Mohammad Sharif Ibrahim Alyahya, Mohamed Amine Amara, Isha Amatya, Junaid Amin, Tarek Tawfik Amin, Saeed Amini, Majid Aminzare, Ganiyu Adeniyi Amusa, Abel Ambaye Anfucho, Yonas Angaw, Abhishek Anil, A.H. Sruthi Anil Kumar, Prapti Anjana, Sanvithi Anjanappa, Samuel Egyakwa Ankomah, Kabilan Annadurai, Aisha Ansari, Mohammad Azam Ansari, Abebe Alemu Anshebo, Boluwatife Stephen Anuoluwa, Iyadunni Adesola Anuoluwa, Saeid Anvari, Sadaf Anwar, Saleha Anwar, Raziq Anwer, Geminn Louis Carace Apostol, Jalal Arabloo, Jesil Mathew Aranjani, Demelash Areda, Abdulfatai Aremu, Olatunde Aremu, Jorge Arias de la Torre, Mahwish Arooj, Harshit Arora, Anton A Artamonov, Francis Arthur-Holmes, Ashokan Arumugam, Nurila Aryntayeva, Muhammad Asaduzzaman, Syed Mohammed Basheeruddin Asdaq, Shewatek Melaku Asefa, Muhammad Sohaib Asghar, Mohammad Ashfaq, Ashraf M Ashmawy, Saad Ashraf, Bernard Kwadwo Yeboah Asiamah-Asare, Moses Asiiimwe, Muhammad Shahzad Aslam, Yuni Asri, Batyrbek Assembekov, Zhupar Atalykova, Seyyed Shamsadin Athari, Samuel Babajide Atoyebi, Yonas Alemu Atra, Sachin R Atre, Abadi Hailay Atsaba, Matteo Augello, Avinash Aujayeb, Zaw Zaw Aung, Sana Javaid Awan, Usman Ayub Awan, Kemal Lemnurowol, Lemessa Assefa A Ayana, Brhane Ayele, Habtamu Molla Ayele, Tigabu Munye Aytenew, Akm Azad, Davood Azadi, Abrahim W Azale, Ayman Azhary, Mohd Yusmaide Aziz, Sadat Abdulla Aziz, Eid MS Azzam, Reem Abass Babiker, Syed Babar Jamal Bacha, Abebe Sorsa Badacho, Muhammad Badar, Huma Bader Ul Ain, Ashish D Badiye, Riadh Badraoui, Emmanuel Badu, Yasemin Bahar, Atif Amin Baig, Mirza Mohammad Ali Baig, Mohammad Hassan Baig, Alpha Umaru Bai-Sesay, Akeem Adebola Bakare, Mohamad Amin Bakhshali, Shankar M Bakkannavar, Mohamed Ibrahim Baklola, Marina Balabekova, Pachamuthu Balakrishnan, Senthilkumar Balakrishnan, Mamadu Baldeh, Yatan Pal Singh Balhara, Olaitan Johnson Balogun, Samuel Alemu Bamboro, Maciej Banach, Ruwanthi Panchali Bandara, Indrajit Banerjee, Biswajit Banik, Palash Chandra Banik, Rajon Banik,

- Megha Raj Banjara, Farhat Bano, Aleksandra Barac, Sanaa Bardaweel, Dennis Bardoe, Hiba Jawdat Barqawi, MD Abu Bashar, Zarrin Basharat, Muhammad Irfan Bashir, Shahid Bashir, Azadeh Bashiri, Prapthi Persis Bathini, Kavita Batra, Ravi Batra, Jagadeesh Bayry, Narasimha M Beeraka, Jina Behjati, Azmeraw Bekele, Andrej Belančić, Suleiman Ayalew Belay, Melesse Belayneh, Samir Bele, Kefyalew T Belete, Uzma Iqbal Belgaumi, Apostolos Beloukas, Samiun Nazrin Bente Kamal Tune, Kenechukwu C Ben-Umeh, Om Prakash Prakash Bera, Robert S Bernstein, Paulo J G Bettencourt, Anmut Endalkachew Bezie, Jeetendra Bhandari, Nikha Bhardwaj, Pankaj Bhardwaj, Ashish Bhargava, Shah Nawaz Ali Bhat, Arushee Bhatnagar, Shuvarthi Bhattacharjee, Sudip Bhattacharya, Ashmin Hari Bhattarai, Gurjit Kaur Bhatti, Jasvinder Singh Bhatti, Bagas Suryo Bintoro, Eshetie Melese Birru, Bijit Biswas, Mohammad Shahangir Biswas, Monirujjaman Biswas, Vishal Biswas, Ahmad Naoras Bitar, Saied Bitaraf, Asnakech Marew Biyadage, Trupti Bodhare, Archith Boloor, Paria Bolourinejad, Samuel Adolf Bosoka, Alejandro Botero Carvajal, Souad Bouaoud, Abdelhakim Bouyahya, Christopher Boxe, Edmond D Brewer, Colin Stewart Brown, Michael Eric Bryan, Sayed Muhammad Ata Ullah Shah Bukhari, Tsion Samuel Bunare, Danilo Buonsenso, Akeem Olayinka Busari, Felix Busch, Yasser Bustanji, Nadeem Shafique Butt, Zahid A Butt, Sravana Deepthi C, Mehtap Çakmak Barsbay, Zhong Cao, Felix Carvalho, Sandra Carvalho, Luca Cegolon, Sinan Cetin, Chiranjib Chakraborty, Sandip Chakraborty, Subhash Chander, Ankit Chandra, Vijay Kumar Chattu, Lam Duc Chau, Anis Ahmad Chaudhary, Sirshendu Chaudhuri, Akhilanand Chaurasia, Snehal Prakash Chavhan, Haiyan Chen, Hana Chen, Junhao Chen, Nicholas WS Chew, Ruixin Chi, Yogapriya Chidambaram, Patrick R Ching, Saravanan Chinnaiyan, Bryan Chong, Deepti Chopra, Hitesh Chopra, Shivani Chopra, Ranjay Kumar Choudhary, ABM Alauddin Chowdhury, Mohiuddin Ahsanul Kabir Chowdhury, Christina Malini Christopher, Devasahayam J Christopher, Sunghyun Chung, Mareli M Claassens, Shiamak Cooper, Alexandru Corlateanu, Claudia Cosma, Natalia Cruz-Martins, Ferran Cuenca-Martínez, Daniela D Vera-Camargo, Omid Dadras, Namrata Dagli, Asrat Berihun Dagnaw, Ephrem Mebratu Dagneu, Lina Dahabiyeh, Hana Mahdi Dahir, Neha Dahiya, Xiaochen Dai, Koustuv Dalal, Ngoc Bao Dang, Samuel Demissie Darcho, Latefa Ali Dardas, Aso Mohammad Darwesh, Sayan Kumar Das, Nihar Ranjan Dash, Manali Datta, Priyadarshini Deb, Ivan Delgado-Enciso, Berihun Assefa Demissie, Muamer Dervišević, Girmay Desalegn, Keshab Deuba, Vinoth Gnana Chellaiyan Devanbu, Devananda Devogoda, Syed Masudur Rahman Dewa, Arkadeep Dhali, Kuldeep Dhama, Premnath Dhanaraj, Amol S Dhane, Sameer Dhingra, Bibha Dhungel, Stefano Di Bella, Marcello Di Pumpo, Daniel Diaz, Igbayilola Yusuf Dimeji, Mbekueh Dinga-Nyoh, Ojas Prakashbhai Doshi, Leila Doshmangir, Vikash Kumar Dubey, Senbagam Duraisamy, Oyewole Christopher Durojaiye, Siddhartha Dutta, Sulagna Dutta, Basil Barth Duwa, Abdelaziz Ed-Dra, Yohannes Tekalegn Efa, Ferry Efendi, Aziz Eftekhari Mehrrabad, Babak Ehsani Zonuz, Matthias M Eikermann, Moaheda E H Eissa, Balisa Mosisa Ejeta, Seraphine Mojoko Eko, Adebayo Abdulateef Ekundayo, Temitope Cyrus Ekundayo, Rabie Adel El Arab, Amine Elbouzidi, Ahmed Eldaboush, Emmanuel Chukwuebuka Elebesunu, Muhammed Elhadi, Yasir Ahmed Mohammed Elhadi, Mohamed Elhoumed, Ashraf A El-Metwally, Mohammad Elraggal, Mohammed Elshaer, Ibrahim Elsohaby, Abdallah Elssir Ahmed, Abdelgawad Salah Eltahawy, Reham Mokhtar ELTarabili, Theophilus I Emeto, Simachew Getaneh Endalamew, Misganu Endriyas, Stanley Chinedu Enah, Muhammad Esa, Setegn Eshetie, Babak Eshtrati, Majid Eslami, Collins Nduka Esomchi, Dean B Everett, Peter Maduabuchi Eze, Obidimma Ezekika, Amanda Ugochi Ezirim, Adeniyi Francis Fagbamigbe, Ayesha Fahim, Ildar Ravisovich Fakhradiyev, Ebenezer Gezahegn Fanta, Bandar M Faqih, Abdullah Farasani, Mohammad Fareed, Aisha Farhana, Syed Muhammad Yousaf Farooq, Munazza Fatima, Nelsensius Klau Fauk, Wisam Fawzi Al-Juhayish, Alireza Feizkhan, Gelana Fekadu, Ginenus Fekadu, Ulrich Membe Femoe, Bikila Regassa Feyisa, Kebede Feyisa, Anteneh Fikrie, Alexander Finnemore, Sofia C Finnemore Capon Diaz, Safia Firdous, Belete Sewasew Firew, Florian Fischer, Federica Fogacci, Marco Fonzo, Carlos Forner Álvarez, Celia Fortuna Rodrigues, Muktar A Gadanya, Márió Gajdács, Adamu Usman Gamawa, Ezhilkugan Ganessane, Ranganath Ganga, Shivaprakash Gangachannaiah, Khadijat Oladayo Ganiyu, Fernando Jr. Barroga Garcia, Alberto L Garcia-Basteiro, William M Gardner, Jacopo Garlasco, Nirmal Gautam, Rupesh K Gautam, Daniel Teshome Gebeyehu, Tsion S Gebre, Yemane Gebremariam Gebre, Haftay Gebremedhin, Mizan Habtemichael Gebresilassie, Nsikakabasi Samuel George, Kalab Yigermal Gete, Kazem Ghaffari, Manal Hadi Ghaffoori Kanaan, Khalid Yaser Ghailan, Ali Ghandili, Ramy Mohamed Ghazy, Sama Ghoba, Ehsan Gholami, Thinh Gia Gia Nguyen, Artyom Urievich Gil, Syed Abdullah Gilani, Syed Amir Gilani, Bikash Ranjan Giri, Alem Abera Girmay, Selam Bogale Gissa, Adrian Eleazar Go, Laszlo Göbölös, Athena Gogali, Awais Gohar, Megha Nishith Gohil, Mahaveer Golechha, Pouya Goleij, Melika Golmohammadi, Wenping Gong, Amador Goodridge, Shirisha Gooty, Jithu Gopakumar, Yitayal Ayalew Goshu, Shi-Yang Guan, Amit Gulati, Kabiru Abubakar Gulma, Bashar Haruna Gulumbe, Bhuvaneshwari Gunasekar, Damitha Asanga Gunawardane, Anil Kumar Gupta, Anish Kumar Gupta, Himanshu Gupta, Sapna Gupta, Sunanda Gupta, Swathi Gurajala, Bruno Gutiérrez, Robert Steven Gutiérrez-Murillo, Jose Guzman-Esquivel, Eduardo Guzmán-Muñoz, Abrahm Tesfaye Habteyes, Awoke Derbie Habteyohannes, Najah R Hadi, Shaymaa Hafedh Alkubaisy, Hideharu Hagiya, Kirubel Tesfaye Hailu, Ramtin Hajibeygi, Pritam Halder, Sobia Ahsan Halim, Souheil Hallit, Raz Mohammed Hama Salih, Mohamed Hamad, Fatimah Othman Hamaraso, Nadia M Hamdy, Mark Hamman, Shadi Hamouri, Ashiru Mohammad Hamza, Didem Han Yekdeş, Asif Hanif, Ashanul Haque, Zitta Barrella Harboe, Arief Hargono, Sara Harsini, Eka Mishbahatul Marah Has, Ahmed I Hasaballah, Md Kamrul Hasan, Shamimul Hasan, Towhid Hasan, Abdul Haseeb, Arezou Hashem Zadeh, Md Saquib Hasnain, Ibrahim Nagmeldin Hassan, Ikrama Hassan, Shafie Abdulkadir Hassan, Omed Hassan Ahmed, Hossein Heli, Yosra A Helmy, Mohamed Fawzi Hemida, Karl Evans Henson, Helal F Hetta, Kamran Heydaryan, Kamal Hezam, Rohman Hikmat, Ramesh Holla, Sadhana Nittur Holla, Nobuyuki Horita, Mehdi Hoseinzadeh, Alamgir Hossain, Lubna Hossain, Md Sabbir Hossain, Md. Jubayer Hossain, Syed Billal Hossain, Mihaela Hostiu, Sorin Hostiu, Mila Nu Nu Htay, Yifei Hu, Junjie Huang, Yongsong Huang, Zhen Huang, Mega Hasanul Huda, Malik Asif Hussain, Md Talib Talib Hussain, Dursa Hussein, Nawfal R Hussein, Mohamed Ibrahim Hussein, Md. Deluwar Hussien, Demilade Olusola Ibironge, Segun Emmanuel Ibitoye, Aminu Alhassan Ibrahim, Khalid S Ibrahim, Anel Ibrayeva, Muhammad Idrees, Adalia Ikiroma, Kevin S Ikuta, Olayinka Stephen Ilesanmi, Irena M Ilic, Milena D Ilic, Aamir Ilyas Abbasi, Salim Ilyasu, Mohammad Tarique Imam, Sakan Binte Imran, Leeberk Raja Inbaraj, Danish Iqbal, Bilal Irfan, Lalu Muhammad Irham, Mustafa Alhaji Isa, Benni Iskandar, Teresa R Iskander, Fahrul Islam, Md Rabiul Islam, Md. Rabiul Islam, Md. Shahinul Islam, Faisal Ismail, Nahlah Elkudssiah Ismail, Wan Ismahana Ismail, Ihoghosa Osamuyi Iyamu, Oluyinka Ajibola Iyiola, Jalil Jaafari, Jaison Jacob, Kathryn H Jacobsen, Venilla Jaganathan, Ayushi Jain, Swati Jaiswal, Mihajlo Jakovljevic, Armaan Jamal, Arshad Jamal, Qazi Mohammad Sajid Jamal, Jazlan Jamaluddin, Jerin James, Safayet Jamil, Syed Sarmad Javadi, Talha Jawaaid, Qassim Jawell Odah Abed, Shubha Jayaram, Yovanthi Anurangi Jayasinghe, Bedilu G Jebena, Jae Joon Jeon, Seongsong Jeong, Nathan T Jibat, Worku Jimma, Akash John, Iain Johnston, Mohammad Jokar, Nabi Jomehzadeh, Pauline Mwenje Jomo, Japhet Haruna Jonah, Jost B Jonas, Jobinse Jose, Alex Joseph, Nitin Joseph, Bhawana Joshi, Krupal Joshi, Farahnaz Joukar, Jacek Jerzy Jozwiak, Hema Shree K, Vaishali K, Zubair Kabir, Vidya Kadashetti, Dler H Hussein Kadir, Mohammad Fahim Kadir, Adel Kadri, Gulgul Kairanbayeva, Ashish Kumar Kakkar, Mehnaz Kamal, Mohammad Azhar Kamal,

Saltanat Kamenova, Oleksandr Kamysnyi, Jiseung Kang, Kehinde Kazeem Kanmodi, Kamala Kannan, Suthanthira Kannan S, Surya Kant, Rami S Kantar, Neeti Kapoor, Paschalis Karakasis, Sajad Karampoor, Indira Karibayeva, Arman Karimi Behnagh, Samad Karkhah, Reddy Praneeth Karnam, Tomasz M Karpiński, Gebrehiwot G Kassa, Nehemia I Kassa, Tsehaynew Kasse, Laura Kassym, Amandeep Kaur, Mohd Adnan Kausar, Awoke Amare Kebede, Fekadu Abera Kebede, Yabets Tesfaye Kebede, Tibebelesassie S Keffie, Varekan Keishing, Chukwudi Keke, John H Kempen, Vikash Ranjan Keshri, Inn Kynn Khaing, Himanshu Khajuria, Anas Khaleel, Asaad Khalid, Sidra Khalid, Hazim O Khalifa, Anees Ahmed Khalil, Ibrahim Khalil, Nouman Khalique, Ahmad Khan, Allahdad Khan, Amjad Khan, Faiz Ullah Khan, Hafiz Nadeem Khan, Hayat Ahmad Khan, Mahvish Khan, Md Abdullah Saeed Khan, Mohd Wajid Ali Khan, Muhammad Arslan Khan, Muhammad Umer Khan, Qamar Uz Zaman Khan, Raees Khan, Ruby Khan, Saif Khan, Shah Alam Khan, Sumaiya Khan, Wasif Ali Khan, Yusuf Saleem Khan, Zahid Khan, Sameer Uttamaro Khasbage, Milad Khorasani, Najmaddin Salih Husen Khoshnaw, Samira Khoshvaght, Pavan Khosla, Sakshi Khurana, Mohsin Khurshid, Hye Jun Kim, Kwanghyun Kim, Min Seo Kim, Yun Jin Kim, Ruth W Kimokoti, Tadele Kinati, Sanjay Kini B, Elizabeth Kassahun Kiros, Teklehaimanot Kiros, Adnan Kisa, Sezer Kisa, Ladli Kishore, Tegene Atamenta Kitaw, Abdul Basith Km, Shivakumar Km, Ko Ko, Sonali Kochhar, Prakash Babu Kodali, Ali-Asghar Kolahi, Joyce Komesoor, Farzad Kompani, Soewarta Kosen, Konstantinos Kostikas, Parvaiz A Koul, Sindhura Lakshmi Koulmane Laxminarayana, James-Paul Kretchy, Kewal Krishan, Bindu Krishnan, Nitya Krishnasamy, Pranav S Kshetriya, Garry Kuan, Burcu Kucuk Bicer, Mohammed Kuddus, Mukhtar Kulimbet, Shweta Kulshreshtha, Emmanuel Kumah, Ashwini Kumar, Chandan Kumar, Dewesh Kumar, Dhasarathi Kumar, Dileep Kumar, Jogender Kumar, Kamal Kumar, Laksh Kumar, Manoj Kumar, Mohan Kumar, Mukesh Kumar, Nithin Kumar, Pardeep Kumar, Pradeep Kumar, Rakesh Kumar, Roshan Kumar, Sandeep Kumar, Sanjay Kirshan Kumar, Sanjeet Kumar, Sumeet Kumar, V Udaya Kumar, Vijay Kumar, Anita Kumari, Setor K Kunutsor, Om P Kurmi, Krishna Prasad Kurpad, Pramod Kumar Kushawaha, Assiya Kussainova, Aria Kusuma, Dian Kusuma, Assylkhan Kuttybayev, Desmond Kuupiel, Ville Kytö, Pallavi L C, Muhammad Awwal Ladan, Leizel P Lagrada, Chandrakant Lahariya, Hanpeng Lai, Timo Lajunen, Balzhan Lakanova, Agung Dwi Laksono, Dharmesh Kumar Lal, Tamás Lantos, Yilma Markos Larebo, Dylan Lasher, Savita Lasrado, Aliyu Lawan, Teniola Lawanson, Thao Thi Thu Le, Yemane Leake, Hayeon Lee, Hoi Leong Lee, Ka Yiu Lee, Sergey Vadimovich Lee, Seung Won Lee, Wei-Chen Lee, Yo Han Lee, Kakithakara Vajravelu Leela, Vasileios Leivaditis, Bati Leta, Si Li, Ambreen Liaqat, Virendra S Ligade, Charles Junior Limula, Ya Lin, Gang Liu, Haipeng Liu, Jue Liu, Xuefeng Liu, Erand Llanaj, Madeeha Shahzad Lodhi, José Francisco López-Gil, Platon D Lopukhov, Arianna Maeve Loreche, Edward Lovering, Jailos Lubinda, Pincheng Luo, Susu Luo, Angelina M Lutambi, Miltiadis D Lytras, Ellina Lytyvak, Hawraz Ibrahim M. Amin, Kevin Sheng-Kai Ma, Zheng Feei Ma, Shamsuddeen Yusuf Ma'aruf, Franciele Maboni Siqueira, Mahmoud Mabrok, Pasquale Maffia, Mohammed Magdy Abd El Razek, Edward Augustine Magwe, Sasikumar Mahalingam, Datchanamoorthy Maheswary, Nozad Hussein Mahmood, Shima Mahmoudi, Azeem Majeed, Ghaseb Naser Makhadmeh, Tabarak Malik, Lesibana Anthony Malinga, Birhanemaskal Malkamu, Biniyam Tedla Mamo, Aseer Manilal, Lokesh Manjani, Farheen Mansoor, Tahir Maqbool, Vijaya Marakala, Basma Hamed Marghani, Francisco Rogerlândio Martins-Melo, J Jenifer Florence Mary, Adeena Maryyum, Roy Rillera Marzo, Sammer Marzouk, Sugeng Mashudi, Clara N Matei, Medha Mathur, Neeta Mathur, Richard James Maude, Sidik Maulana, Md. Sabbir Ahmed Ahmed Mayen, Prosanjit Mazumder, John Robert Carabeo Medina, Jitendra Meena, Sandeep K Meharwal, Fahad Mehmood, Subhash Mehto, Tesfahun Mekene Meto, Agezege Asegid Mekonnen, Addisu Melese, Ziad Ahmed Memish, Walter Mendoza, Godfred Antony Menezes, Ritesh G Menezes, Emiru Ayalew Mengistie, Michelangelo Mercogliano, Tarig H Merghani, Muayad Aghali Merza, Rahwa Adisu Mesfin, Tomislav Mestrovic, Chamila Dinushi Kukulege Mettananda, Mohamed M M Metwally, Sandrine Donfack D Mewoabi, Yonas Abebe Mezemir, Turnwait Otu Michael, Andrea Michelerio, Giuseppe Minervini, Long Chiau Ming, Gk Mini, Shahzad Mirza, Bhagyashree Mishra, Vinaytosh Mishra, Arup Kumar Misra, Babak Moazen, Dhruti Modi, Farduus Ibraahim Mohamed, Fatima Awad Elkarim Elfaki Mohamed, Hebatalla Mohamed, Mona Gamal Mohamed, Nouh Saad Mohamed, Khabab Abbasher Hussien Mohamed Ahmed, Taj Mohammad, Mohammad Reza Mohammadi, Hussien Mohammed, Mustapha Mohammed, Shafiu Mohammed, Yahaya Mohammed, Syam Mohan, Yugal Kishore Mohanta, Aroop Mohanty, Khayriyyah Mohd Hanafiah, Mohammad Mohseni, Afrasim Moin, Ali H Mokdad, Mohammad Ali Moni, Paula Moraga, Anthony Kwame Morgan, Mahdis Morovvati, Reza Mosaddeghi Heris, Freiser Eceomo Cruz Mosquera, Md. Shihab Mostafa, Rohith Motappa, Amin Mousavi Khaneghah, Mohamed Awad Abdalaziz Mousnad, Hagar L Mowafy, Ahmed Msherghi, Rabia Mubarak, Sumaira Mubarik, Kindie Fentahun Muchie, Steward Mudenda, Shiv Kumar Mudgal, Jibran Sualeh Muhammad, Oscar J Mujica, Amartya Mukhopadhyay, Francesk Mulita, Malaisamy Muniyandi, Kavita Munjal, Scherko Hama Law Murad, Kamarul Imran Musa, Sani Musa, Abdissalam Hassan Hassan Muse, David Musoke, Ghulam Mustafa, Ghulam Mustafa, Adam Mustapha, Saravanan Muthupandian, Raman Muthusamy, Claude Mambo Muvunyi, Lutfi Muzaqi, Syrsulu Myrzagulova, Woojae Myung, Ahamarshan Jayaraman Nagarajan, Karikalan Nagarajan, Ganesh R Naik, Firzan Nainu, Sanjeev Nair, Mohammad Zeeshan Najm Najm, Gopal Nambi, Vinay Nangia, Ibrahim A Naqid, Sreenivas Narasimha Swamy, Shumaila Nargus, Abdulqadir J Nashwan, Mahmoud Nassar, Zuhair S Natto, Javid Nauman, Zakira Naureen, Samidi Nirasha Kumari Navaratna, Biswa Prakash Nayak, Kinjal Nayak, Javad Nazari, G. Takop Nchanji, Robert Ndege, Pacifique Ndishimye, Nini Asfaw Negash, Chakib Nejari, Samata Nepal, Jean Claude Semuto Ngabonziza, Cuong Tat Nguyen, Duy Cao Nguyen, Hau Thi Hien Nguyen, Nghia Phu Nguyen, Tham Thi Nguyen, Thanh Van Nguyen, Quan Nguyen Khoi, Minh Thien Nguyen Thien, Ambe Marius Ngwa, Robina Khan Niazi, Dina Nur Anggraini Ningrum, Vikram Niranjan, Kottakkar Noppo Nisar, Richmond Nketia, Chukwudi A Nnaji, Lawrence Achilles Nnyanzi, Shuhei Nomura, Ali Aahil Noorali, Sana Noreen, Emira Noumi, Mohammed Noushad, Chisom Adaobi Nri-Ezedi, Jean Claude Nshimiyimana, Mpiko Ntsekhe, Fred Nugen, Mengistu H Nunemo, Nurfatimah Nurfatimah, Ahmad Dzulfikri Nurhan, Eze Kingsley Nwangwa, Chimezie Igwegbe Nzopotam, Ogochukwu Janet Nzopotam, Bogdan Oancea, George Obaïdo, Vincent Adeiza Obakachi, Kwabena Obiri - Yeboah, Saheed Adeyinka Odediji, Alex Odoom, Michael Safo Oduro, Akinyemi O D Ofakunrin, Felix Akpojene Ogbo, Anthony Jide Ogunbadewa, Olawale Ogundiran, In-Hwan Oh, Rasaq A Ojasanya, Tolulope R Ojo-Akosile, Debra Ukamaka Okeh, Akinkunmi Paul Okekunle, Olalekan John Okesanya, Osaretin Christabel Okonji, Lenz Nwachinemere Okoro, Oluwaseyi Isaiah Olabisi, Musa Kolapo Oladejo, Oladotun Victor Olalusi, David Bamidele Olawade, Antonio Olivas-Martinez, Abdulhakeem Abayomi Olorukooba, Adetayo Olorunlana, Ahmed Olowo-okere, Olutosin Samuel Olusola, Ruth Toluwani Olusola, Yinka Doris Oluwafemi, Vivian N Onaemo, Francisca Ogochukwu Onukansi, Franklyn Opara, Aksoltan Shyhdurdevna Oradova, Edgar Ortiz-Brizuela, Godwin Osawaru Osaigbovo, Eric Osei, Uchechukwu Levi Osuagwu, Chinenye Oche Otokpa, Oche Joseph Otokpa, Jerry John Ouner, Amel Ouyahia, Tope Oyelade, Kehinde Adewole Oyeniran, Ilker Ozsahin, Nagaraj P, Mahesh P A, Jagadish Rao Padubidri, Dimpal Manilal Paija, Vijayakumar Palaniswamy, Tejasri Paluvai, Alok Kumar Panda, Jibanjyoti Panda, Sujogya Kumar Panda, Songhomitra Panda-Jonas, Sadanand Pandey, Seithikurippu R Pandi-Perumal, Georgios D Panos, Ioannis Pantazopoulos, Ilias Papadimopoulos, Shahina Pardhan, Amrita Parida, Romil R Parikh, Arpit Parmar, Pragnesh Parmar, Swapnil Parve, Sameena Parveen, Maja Pasovic, Roberto Passera,

- Dorababu Patchva, Kunal Nitinkumar Patel, Mehali Lalitbhai Patel, Mitesh Patel, Niki Rut Patel, Satyananda Patel, Shivani Patel, Vyoma Nitinkumar Patel, Aslam Ramjan Pathan, Vipin Patidar, Shankargouda Patil, Mohamed Anas Patni, Shrikant Pawar, Arun Mitra Peddireddy, Saikiran Pendem, Flavia Pennisi, Veincent Christian Filipino Pepito, Prince Peprah, João Perdigão, Norberto Perico, Ionela-Roxana Petcu, Olumuyiwa James Peter, Tung Thanh Pham, Zayar Phyto, Edoardo Pirera, Roman V Polibin, Thantrira Porntavetus, Maarten J Postma, Farzad Pourghazi, Attur Ravindra Prabhu, Rifky Octavia Pradipta, Nk Prasanna, Michal Pruc, Dhiren Punja, Muhammad Usman Qamar, Husam Qanash, Nameer Hashim Qasim, Xiang Qi, Nuzul Qur'aniati, Shahazad Niwazi Qurashi, Ali Qureshi, Safia Obaidur Rab, Navid Rabiee, Brian Eka Rachman, Igor Radenkov, Venkatraman Radhakrishnan, Yashpal Singh Raghav, Pracheth Raghuveer, Sheu Kadiri Rahamon, Fakher Rahim, Sajjad Rahimi, Mahbubur Rahman, Md Mosfequr Rahman, Mohammad Hifz Ur Rahman, Mohammad Meshbahur Rahman, Mosiur Rahman, Muhammad Aziz Rahman, Amir Masoud Rahmani, Setyaningrum Rahmawaty, Hakim Rahmoune, Amita Rai, Sunil Kumar Raina, Jeffrey Pradeep Raj, Gunaseelan Rajendran, Judah Rajendran, Vinoth Rajendran, Humam Rajha, Nurgali Amanbaiuly Rakhymbayev, Rafat Rakoczy, Vignesh Ramachandran, Hazem Ramadan, Mahmoud Mohammed Ramadan, Kadar Ramadhan, Chitra Ramasamy, Shakthi Kumaran Ramasamy, Balaji Ramraj, Rishabh Kumar Rana, Chhabhi Lal Ranabhat, Nemanja Rancic, Smitha Rani, Asad Gul Rao, Kumuda Rao, Mithun Rao, Rajath Rao, Sowmya J Rao, Davide Rasella, Ashkan Rasouli-Saravani, Prateek Rastogi, Rama Shankar Rath, Isha Rathi, Devarajan Rathish, Santosh Kumar Rauniyar, Lokesh Ravi, Renju Ravi, Sujith Sri Surya Ravi, Mudalsha Ravina, Salman Rawaf, Karthik Rayapureddi, Muhammad Liaquat Raza, Elrashdy M Redwan, Wajiha Rehman, Bhagwan Narayan Rekadwad, Giuseppe Remuzzi, Kannan RR Rengasamy, Luis Felipe Reyes, Mina Rezaei, Nazila Rezaei, Nima Rezaei, Mohsen Rezaeian, Omid Rezahosseini, Yohanes Andy Rias, Moattar Raza Rizvi, Syed Mohd Danish Rizvi, Erna Rochmawati, Jefferson Antonio Buendia Rodriguez, Ravi Rohilla, Chanapong Rojanaworarat, Debby Syahrul Romadlon, Moustaq Karim Khan Rony, Kevin T Root, Allen Guy Patrick Ross, Kunle Rotimi, Himanshu Sekhar Rout, Parimal Roy, Priyanka Roy, Sharmista Roy, Shubhanjali Roy, Guilherme de Andrade Ruela, Tanko Rufai, Godfrey Mutashambara Rwegerera, Devarajulu Reddy S, Jaya Lakshmi S S, Aly M A Saad, Khaled Saad, Sura Saad Abdullah, Seyed Kiarash Sadat Rafiei, Muhammad Nabeel Saddique, Ehsan Sadeghi, Bassem Sadek, Halima Sadia, Mohammad Reza Saeb, Mohd Saeed, Rida Fatima Fatima Saeed, Umar Saeed, Mehdi Safari, Sher Zaman Safi, Muhammad Safiullah, Amene Saghazadeh, Ashok Kumar Sah, Indranil Saha, Fatemeh Saheb Sharif-Askari, Narjes Saheb Sharif-Askari, Amirhossein Sahebkar, Kashif Iqbal Sahibzada, Vaibhav Sahni, Alaka Sahoo, Ambika Sahoo, Binaya Krushna Sahu, Muhammad Soaib Said, S Mohammad Sajadi, Morteza Saki, Noha Salah Salah Soliman, Luciane B Salaroli, Muhammad Zubair Saleem, Rahman Shah Zaib Saleem, Zikria Saleem, Mohamed A Saleh, Amir Salek Farrokhi, Ramatu Salisu, Janmejaya Samal, Yoseph Leonardo Samodra, Fasineh Shekuba Samura, Rama Krishna Sanjeev, Sathish Sankar, Bruno Piassi Sao Jose, Siddaram Shivaji Sarate, Aswini Saravanan, Hemen Sarma, Mohammad Sarmadi, Gargi Sachin Sarode, Sachin C Sarode, Benn Sartorius, Michele Sassano, Murtaja Sate, Mukesh Kumar Sathya Narayanan, Maheswar Satpathy, Naghmeh Sattarahmady, Shakta Mani Satyam, Amar Saurabh, Mehrdad Savabi Far, Tual Sawn Sawn Khai, Sangeeta Gopal Saxena, Neil Schluger, Art Schuermans, Mohammed Assen Seid, Durairaj Sekar, Siddharthan Selvaraj, Yuliya Semenova, Pallav Sengupta, Subramanian Senthilkumaran, Yashendra Sethi, Ferit Sevim, Dina Seyed, Allen Seylani, Ahmed Nabil Shaaban, Arash Shaban-Nejad, Aroosh Shabbir, Muhammad Ashir Shafique, Disha Sanjivkumar Shah, Liloma Shah, Sajid Ali Shah, Syed Ali Raza Shah, Shazlin Shaharudin, Samiah Shahid, Syed Ahsan Shahid, Endrit Shahini, K M Shahunja, Moyad Jamal Shahwan, Babar Tasneem Shaikh, Masood Ali Shaikh, Sunder Sham, Anas Shamsi, Alfiya Shamsutdinova, Siva Kumar Shanmugam, Amit Baran Sharangi, Nadim Sharif, Sumaira Sharif, Amin Sharifan, Fariborz Sharifian Jazi, Avimanu Sharma, Bhoopesh Kumar Sharma, Kamlesh Sharma, Manoj Sharma, Nidhi Sharma, Ravi Kumar Sharma, Sourabh Sharma, Vandana Sharma, Rajesh P Shastri, Ramzi Shawahna, Samrawit Shawel, Babangida Shehu Bappah, Somia Shehzadi, Ali Sheidaei, Nishat Ahmed Sheikh, Suchitra M Shenoy, Samendra P Sherchan, Subuhi Sherwani, Mahabalesh Shetty, Pavanchand Shetty, Premalatha K Shetty, Ranjitha S Shetty, Suraj S Shetty, Robert Adamu Shey, JiaHao Shi, Muhammad J A Shiddiky, Desalegn Shiferaw, Md Monir Hossain Shimul, Arun Shirali, Priyanka Arun Shirali, Reza Shirkooi, Aminu Shittu, Birhanu Feleke Shitu, Km Shivangi, Velizar Shivarov, Ambreen Shoaib, Deborah Oluwaseun Shomuyiwa, Sina Shool, Seyed Afshin Shorofi, Sunil Shrestha, Dharmasheel Shrivastav, Suleiman Adeiza Shuaibu, Nicole Remaliah Samantha Sibuyi, Abu Bakkar Siddique, Samra Siddiqui, Luis Manuel Lopes Rodrigues Silva, Aditya Pratap Singh, Akanksha Singh, Amit Singh, Baljinder Singh, Harmanjit Singh, Harpreet Singh, Jasvinder A Singh, Jawahar Singh, Kalpana Singh, Madhusudan Prasad Singh, Paramdeep Singh, Poornima Suryanath Singh, Prince Kumar Singh, Rakesh K Singh, Samer Singh, Satwinder Singh, Siddharth Singh, Surendra Singh, Surjit Singh, Mukesh Kumar Sinha, Ratnesh Sinha, Robert Sinto, Kosala Sirisena, Indra Sivakumar, Gisha Sivan, Mejd Snoussi, Farrukh Sobia, Md.Salman Sohail, Solikhah Solikhah, Ahmed M Soliman, Yonatan Solomon, Prashant Sood, Anna Maria Spagnolo, Lokasani Vijaya Sree, Saraswathy Sreeram, Chandrashekhar T Sreeramreddy, Kannan Sridharan, Vignes Anand Srinivasulu, Shyamkumar Sriram, Muhammad Haroon Stanikzai, Paschalis Steiropoulos, Brendon Stubbs, Chen-Yang Su, Balachandran Subramanian, Meenakshi Sundari Subramaniam Natarajan, Agus Sudaryanto, Chee Fai Sui, Sahabi K Sulaiman, Muritala Suleiman Odidi, R Endro Sulistyo, Mark J M Sullman, Marjia Sultana, Zhong Sun, Thanigaivel Sundaram, Krishna Mohan Surapaneni, Jithin Surendran, Suryanti Suryanti, Tarun Kumar Suvvari, Chandan Kumar Swain, Lukasz Szarpak, Ramin Tabibi, Mohammad Tabish, Biniyam Beyene Tabor, Rebecca Tadokera, Balamurugan Tagiuran, Zanan Mohammed-Ameen Taha, Amir Tahavvori, Niloofar Taheri, Shima Tajabadi, Iman M Talaat, Avni Tambi, Mircea Tampa, Ephrem Tamrat, Ching Siang Tan, Jianye Tan, Mohsan Tanveer, Mengistie Kassahun Tariku, Saba Tariq, Elvis Enowbeyang Tarkang, Noshin Anjum Tasmi, Aigul Yelgondiyevna Tazhiyeva, Mebrahtu G Tedla, Faniel Alemayehu Tefera, Nebyu Lakew Tefera, Abainash Tekola, Reem Temsah, Tigabu Kidie Tesfie, Azimeraw Arega Tesfu, Belay Tessema, Vidhi Thakar, Chandan Kumar Thakur, Rekha Thapar, Muthu Thiruvengadam, Rekha Thiruvengadam, Christina S Thornton, Jansje Henny Vera Ticoalu, Madi Tleshev, Enyu Tong, Arole Darwin Touko, Khai Hoan Tram, Quynh Thuy Huong Tran, Nguyen Tran Minh Duc, Domenico Trico, Indang Trihandini, Tulika Tripathi, Vishwas Tripathi, Jaya Prasad Tripathy, Sarvodaya Tripathy, Anastasia Troia, Claudia Truppa, Gary Tse, Mike Tuffour Amirikah, Lawrence Sena Tuglo, Santhosh Kumar Tumkur Narayanappa, Munkhtuya Tumurkhuu, Biruk Shalmeno Tusa, Md Jamal Uddin, Atta Ullah, Himayat Ullah, Riaz Ullah, Saeed Ullah, Srikanth Umakanthan, Eduardo A Undurraga, Brigid Unim, Era Upadhyay, Jibrin Sammani Usman, Chiamaka Henrietta Uwalaka, Olivier Uwishema, Sameer Uz Zaman, Dilber Uzun Ozsahin, Hande Uzunçibuk, Rakesh V R, Asokan Govindaraj Vaithinathan, Mathew Joseph Valamparampil, Raman Swathy Vaman, Joe Varghese, Levin Jake Varghese, Vishnu Priya Veeraraghavan, Anoop Velayudhan, U Venkatesh, Poonam Verma, Giacomo Pietro Vigezzi, Vignesh Vikram, Simone Villa, Tuan Vinh, Antonio Vitiello, Vasily Vlassov, Linh Vu, Thinh Toan Vu, Abubakar Sadiq Wada, Rashim Wadhwa, Fasil Wagnew, Yasir Waheed, Agnes Wamuyu Wamai, Eric Lee Wan, Liang Wang, Xuemei Wang, Yanzhong Wang, Tanveer A A Wani, Ahmed Bilal Waqar, Anthony Ike Wegborn, Desalegn Desalegn Weldebrhan, Yonas Gebregziabher Weldegerima, Shimels Getaneh Weldemedhn, Dakshitha Praneeth Wickramasinghe,

Nuwan Darshana Wickramasinghe, Angga Wilandika, Sri Winarni, Andrew Awuah Wireko, Frankline Sevidzem Wirsyi, Fikirt Woldeeslassie Woldeyohannes, Tewodros Eshete Wonde, Yohannes Chemere Wondmeneh, Yen Jun Wong, Florence Gyembuzie Wongnaah, Shulan Wu, Zenghong Wu, Tadele Kassahun Wudu, Ratna Dwi Wulandari, Mukesh Kumar Yadav, Vikas Yadav, Sajad Yaghoubi, Saba Yahoo, Galal Yahya, Rebaz Othman Yahya, Yusuf Yakubu, Natsuhiko Yamada, Chengliang Yang, Haibo Yang, Hanwen Yang, Kivanc Yangi, Mohamed A Yassin, Thomas Kidanemariam Yewodaiw, Saber Yezli, Siyan Yi, Xinglin Yi, Yazachew Engida Yismaw, Dong Keon Yon, Naohiro Yonemoto, Marcel Yotebieng, Abdilahi Yousuf, Monal Yuwanati, Mubashir Zafar, Manijeh Zaghampour, Fathiah Zakham, Nelson Zamora, Iman Zare, Seema Zargar, Michael Zastrozhin, Saad Zbiri, Mohammed G M Zeariya, Jehan Zeb, Alemu Birara Zemariam, Yohannes Zenebe, Basit Zeshan, Daniel Xin Zhang, Haijun Zhang, Julio Min Fei Zhang, Shun-Xian Zhang, Yue Zhang, Yunquan Zhang, Zhichang Zhang, Murat Zhanuzakov, Chenyu Zhao, Jianhui Zhao, David X Zheng, Lei Zheng, Bin Zhu, Abzal Zhumagaliuly, Hafsa Zia, Magdalena Zielińska, Alimuddin Zumla, Ahed H Zyouid, Sa'ed H Zyouid, Shaher H Zyouid, Theo Vos, Simon I Hay, Mohsen Naghavi, Marie Ng**, Christopher J L Murray**, Hmwe Hmwe Kyu**.

*Joint first authors.

**Joint senior authors.

Affiliations

For the affiliations of individual authors, please see appendix 3 (pp 9–49).

Contributors

Please see appendix 3 (pp 49–68) for more detailed information about individual author contributions to the research, divided into the following categories: managing the overall research enterprise; writing the first draft of the manuscript; primary responsibility for applying analytical methods to produce estimates; primary responsibility for seeking, cataloguing, extracting, or cleaning data; designing or coding figures and tables; providing data or critical feedback on data sources; developing methods or computational machinery; providing critical feedback on methods or results; drafting the manuscript or revising it critically for important intellectual content; and managing the estimation or publications process. Members of the core research team for this topic area had full access to the underlying data used to generate estimates presented in this Article. All other authors had access to and reviewed estimates as part of the research evaluation process, which includes additional stages of formal review.

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Data sharing

The statistical code used for GBD estimation is available upon request; requests should be sent to engage@healthdata.org. To download the data used in these analyses, please visit the Global Health Data Exchange website at <https://ghdx.healthdata.org/gbd-2023> and <https://ghdx.healthdata.org/record/ihme-data/gbd-2023-tuberculosis-1990-2023>.

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